Fitting ω Centauri with a compact action-based distribution function

Model, data issues, fitting performance and the status of the dark-matter question

oCen_dm project notes

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Abstract

We fit the surface-brightness profile and 89 binned velocity-dispersion measurements of ω Centauri (HST and Gaia EDR3 proper motions, MUSE line-of-sight velocities) with a self-consistent spherical equilibrium. Its stars follow a positive, regularized exponential action distribution function (DF); stellar remnants and an optional cored dark-matter (DM) halo enter as prescribed densities. The original DF, with a single orbital-anisotropy transition, cannot fit the combined data: the best fit has $\chi^2_{\rm kin}/N = 4.80$. All valid starts converge to the same branch minima, so the optimizer is not the cause. Wider search bounds, free distance and the HST/Gaia boundary are also excluded. The model fits HST+MUSE alone ($\chi^2_{\rm kin}/N = 1.43$) and Gaia alone (0.88), but these subsets demand opposite anisotropy: radial at small actions, tangential at large actions. A second anisotropy transition, already part of the model definition, removes most of the conflict. Without a halo it reaches $\chi^2_{\rm kin} = 159.2$ for 89 bins ($\chi^2_{\rm kin}/N = 1.79$), better than the best existing Jeans fit to the same bins ($\chi^2 = 190.0$). Its intrinsic anisotropy reproduces the independently inferred Jeans turnover out to ~ 30 pc. Remaining misfits are localized: the outer edge of the MUSE field and the tangential proper motions at 4–20 pc. Adding a free cored halo lowers the objective by 31 but drives two coordinates to their bounds and rearranges the anisotropy. This is a mass–anisotropy trade, not a DM measurement. We then replace the adopted-error photometric profile by a Poisson likelihood for star counts (HST $F625W < 19$ inside 250 arcsec, Gaia $G < 17$ outside) and fit the distance with its literature prior. The kinematic fit is unchanged in quality ($\chi^2_{\rm kin}/N = 1.78$, no dataset above 2.4), the outer anisotropy that the old weighting had imposed disappears, and the star counts become the worst-fitted term (HST deviance 4.9 per bin). With this likelihood a free halo converges to $\rho_{20} = 0$, and a profile likelihood over fixed ρ_{20} rises monotonically ($\Delta = +2.2$ at $\rho_{20} = 1$, $+10.6$ at $4\,{\rm M}_\odot\,{\rm pc}^{-3}$), driven by the kinematics. Twelve realistic mocks show, however, that ρ_{20} is recovered with a scatter of 0.5–0.9 ${\rm M}_\odot\,{\rm pc}^{-3}$ when the halo scale is free, as large as the observed rise, while the total mass profile is recovered to a few per cent over 2–40 pc. The data therefore determine the mass profile but not yet the stellar/DM split. An external stellar mass from the oMEGACat present-day mass function (MIST isochrones) gives 8.3–9.8 ${\rm M}_\odot$ of luminous mass per counted $F625W < 19$ star in the kinematically relevant zones, 12–27 per cent below the dynamical 10.67. Reported as a function of this mass per star, the halo switches on below the dynamical value ($\rho_{20} = 2.6$ at 9.0, 4.3 at 8.0); mocks show that a 16 per cent underestimate produces exactly such a halo on a no-DM truth, so the photometric budget must reach ~ 3 per cent before it can decide. With the stellar mass pinned the residual degeneracy is remnants against a compact halo, at the 1 per cent level of the enclosed mass. Finally, N-body realizations of a no-DM and a DM model (equilibrium verified in isolation) were disrupted on ω Cen's orbit for 1.96 Gyr in a static Milky Way. Their tails differ by at most ~ 30 per cent in the observable quantities (tail-to-cluster star ratio, near-tail dispersion); the mismatches with the observed tails are common to both models and point to the orbit, integration time and missing rotation rather than to DM.

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1 Aims

The project has two objectives, reported separately throughout:

1. **Fit the data well**: find a physically admissible equilibrium model that reproduces the light profile and all kinematic profiles within their uncertainties.
2. **Place meaningful constraints on the DM content**, chiefly the DM density at 20 pc, ρ_{20}.

The two objectives pull against each other. A more flexible stellar DF fits better, but it can also mimic a halo, which weakens the DM constraint. A model must pass objective 1 before any DM statement from it is meaningful. It should also be no more flexible than the data require.

This note covers the observed-data fits of 23–24 September 2026, the mass-function constraint and the N-body tidal-disruption runs of 24–26 September (Sections 6.4–7). The model definition is in `docs/compact_df_model.tex`, the implementation tests in `docs/REGULARIZED_DF_IMPLEMENTATION.md`, and the full log in `JOURNAL.md`.

2 Data

2.1 Datasets

All fits use one frozen data snapshot (fingerprint `37dfc7fa...`, file `results/df/pilot_no_dm_20260922/`). The same snapshot was used by the earlier Jeans fits, so their χ^2 values are directly comparable. Radii are converted at 0.02633 pc per arcsec, i.e. $D = 5.43$ kpc (the project's adopted distance, from Baumgardt and Vasiliev, 2021).

Table 1: The observational inputs. R gives the range of bin centres, with bin edges in parentheses. "Rel. err" is the range of the published dispersion uncertainty relative to the dispersion. $\bar{v}^2/\sigma^2$ is the largest ratio of the adopted mean-streaming term to the squared dispersion (Section 2.2).

Dataset	n	R [pc]	Rel. err	max $\bar{v}^2/\sigma^2$	Provenance
HST PM radial	21	0.10–9.0 (0.09–9.5)	0.2–7.1%	0	Our measurement, `selection_hq_astrometry`, cluster+field mixture, to 360 arcsec
HST PM tangential	21	0.10–9.0 (0.09–9.5)	0.3–7.1%	0.25	As above; the Vasiliev and Baumgardt (2021) rotation curve is added to the model
MUSE LOS	29	0.15–7.6 (0–8.7)	1.4–11%	0.14	oMEGACat VI adaptive bins (Häberle et al., 2025), dispersion about the fitted rotation curve
Gaia EDR3 PM radial	9	9.4–56.6 (7.9–63.2)	1.4–10%	0	Our measurement, one component, raw errors
Gaia EDR3 PM tangential	9	9.4–56.6 (7.9–63.2)	1.9–10%	0.38	As above, published rotation
Surface brightness	82	0.045–68.1	0.10–0.58 mag	–	oMEGACat star counts ($F625W < 19$, 129,074 stars) inside 25 arcsec, Trager et al. (1995) V band outside

The kinematic likelihood uses the published asymmetric uncertainties in a split-normal χ^2. For bin i, with model m_i, datum d_i and errors $(\sigma_{i,-}, \sigma_{i,+})$,

$$\chi^2_{\rm kin} = \sum_i \left(\frac{m_i - d_i}{\sigma_{i,\pm}} \right)^2 , \qquad \sigma_{i,\pm} = \begin{cases} \sigma_{i,+}, & m_i > d_i \\ \sigma_{i,-}, & m_i \le d_i \end{cases} . \tag{1}$$

Model predictions are averaged over each bin's saved radial selection nodes, in the units of the data. The photometric term uses a single profiled magnitude zero point (constant stellar mass-to-light ratio) and the adopted magnitude errors. The fitted objective is $\chi^2_{\rm kin} + \chi^2_{\rm phot}$.

2.2 Data issues

The following properties of the data limit how the goodness of fit can be interpreted. None of them was altered in the fits reported here: all fits use the published errors (Section 8).

Very small formal kinematic errors. The HST PM dispersions reach 0.2–0.3% precision (e.g. $0.592 \pm 0.001\,\mathrm{mas\,yr^{-1}}$ at 7.1 pc). The Gaia dispersions have 1–2% errors (0.005–$0.007\,\mathrm{mas\,yr^{-1}}$) at 17–35 pc. The Gaia errors are purely statistical. Gaia EDR3 has spatially correlated astrometric systematics on small angular scales and underestimated per-star uncertainties in crowded fields (Lindegren et al., 2021; Vasiliev and Baumgardt, 2021). If per-star errors ϵ are underestimated, the deconvolved dispersion $\sigma^2_{\rm true} = \sigma^2_{\rm obs} - \epsilon^2$ is biased high. The effect is a bias in the measurement, not only in its error bar. At this precision, small calibration effects can dominate χ^2. We have not re-derived either profile with a calibrated error model. That is an open, separate data task.

Heterogeneous photometry. The photometric profile splices oMEGACat star counts (inside 25 arcsec, zero point offset by +16.991 mag over 30–100 arcsec) with the Trager et al. (1995) compilation. Its errors are adopted weights, $0.1\,\mathrm{mag}/\sqrt{w}$, not measured uncertainties; the splice covariance is omitted. Ten pairs of full-weight (0.1 mag) points at the same radius differ by up to 0.40 mag, i.e. 0.13 mag per point. A perfect model would therefore give $\chi^2_{\rm phot}/N \simeq 1.7$ on full-weight points (an estimate from ten pairs). Section 5.5 shows that the photometric misfit comes mainly from points of opposite sign at the same radii at 10–40 pc.

Rotation is subtracted, not modelled. The DF is non-rotating (even in L_z). The published dispersions are measured about a mean motion, so each model second moment is corrected as $\sigma_{\rm model} = \sqrt{\langle v^2\rangle - \bar{v}^2}$. For MUSE this uses $\bar{v}^2 = v^2_{\rm rot}/2$ from the fitted rotation curve; for the tangential PMs it uses the Vasiliev and Baumgardt (2021) curve. Adding an odd-L_z part to an even DF leaves the second moments unchanged. The subtraction is therefore exact for that model class, provided the measured dispersion is taken about the correct local mean. The approximations are geometric: the annular average, inclination, and sphericity. MUSE also reveals a counter-rotating core (Pechetti et al., 2024). The correction is largest exactly where the fits are worst: $\bar{v}^2/\sigma^2$ reaches 0.38 in the Gaia tangential bins and 0.14 at the MUSE outer edge.

Correlated bins. The HST radial and tangential profiles are built from the same stars, as are the two Gaia components. Treating bins as independent makes absolute χ^2/N values and p-values indicative only.

The HST/Gaia boundary. The HST bins end at 9.5 pc and the Gaia bins begin at 7.9 pc (bin edges). In the one near-coincident pair of bins (9.0 and 9.4 pc) the surveys agree within errors: $\sigma_T = 0.465 \pm 0.027$ vs $0.474 \pm 0.049\,\mathrm{mas\,yr^{-1}}$ and $\sigma_R = 0.531 \pm 0.029$ vs $0.510 \pm 0.052\,\mathrm{mas\,yr^{-1}}$. Both errors are large, so this is only a weak consistency check.

Fixed distance. $D = 5.43$ kpc in Sections 5.1–5.7; free with a Gaussian prior from Section 5.6 on.

2.3 Star counts as the tracer-density constraint

The adopted-error magnitude profile has no sampling distribution, so its weight relative to the kinematics was arbitrary (it made up $\sim 60\%$ of the objective in Sections 5.3–5.4). From Section 5.6 on it is replaced by binned star counts with a Poisson likelihood and no adopted errors:

- **HST**: all oMEGACat stars with a measured $F625W < 19$ (117,751 stars) in 20 logarithmic annuli from 2 to 250 arcsec about the catalogue's pixel centre. The authors' photometric quality flag is *not* applied: the fraction of $F625W < 19$ stars carrying it rises from 0.81–0.83 at 8–22 arcsec to 0.94–0.95 beyond 175 arcsec, which would imprint a 12–15% radial incompleteness. Without it the ratio $N(F625W < 20)/N(F625W < 19)$ is 2.0–2.1 in every annulus, so the bright sample is complete. Footprint coverage per annulus (from the occupancy of 3-arcsec cells) is ≥ 0.956; the bins stop at 250 arcsec, where the mosaic edge begins. Field contamination is below 0.5%, so no field term.
- **Gaia EDR3**: all stars in the Vasiliev and Baumgardt (2021) field catalogue with $G < 17$ (18,289 stars, members and field), in 12 annuli from 300 to 2400 arcsec. $N(G < 16)/N(G < 15)$ and $N(G < 17)/N(G < 16)$ are flat (1.8–2.0) from 300 arcsec outward, whereas $G < 18$ is depleted inside ~ 500 arcsec by crowding, hence the cut. A uniform field density is fitted; membership probabilities are not used.

For bin i with area A_i (coverage times annulus area) the model predicts $\mu_i = A_i\,(a\,\bar{\Sigma}_i + b)$, with $\bar{\Sigma}_i$ the model surface density averaged over eight equal-count radial nodes of the counted stars, $a \geq 0$ the number of stars per unit model surface density and $b \geq 0$ the field density (Gaia only), both profiled at every evaluation by maximizing the Poisson likelihood. The residual vector uses signed deviance residuals, whose squared sum is the Poisson deviance $D = 2\sum_i [N_i \ln(N_i/\mu_i) - (N_i - \mu_i)]$, so least squares maximizes the Poisson likelihood. The two selections sample different stellar populations (faint HST stars, bright Gaia giants); each has its own amplitude, and the common shape is the mass-follows-light assumption. The Trager compilation is no longer used. Script: `bin/build_count_profiles.py`.

3 Model

3.1 Stellar distribution function

Following `compact_df_model.tex` (Section 2), the stars follow a positive DF of the radial action J_r and total angular momentum $L = J_z + |J_\phi|$ in the total potential. Constant-DF contours are obtained by integrating

$$\frac{\mathrm{d}L}{\mathrm{d}J_r} = -g(J_r, L), \qquad g = g_H(J_r, L)\exp\left[-b(q)\sin\left(\frac{\pi c}{2}\right)\right], \qquad q = J_r + L, \quad c = \frac{L}{q}, \tag{2}$$

to the L axis, which defines a label $L_c(J_r, L)$. The DF is

$$f_\star(J_r, L) = A_\star \exp\left[-\left(\frac{L_c(J_r, L)}{J_0}\right)^\alpha\right], \qquad A_\star = \frac{M_\star}{(2\pi)^3 \mathcal{I}_\star}, \tag{3}$$

where $\mathcal{I}_\star$ is the numerically converged action-space normalization. The reference $g_H = \Omega_r/\Omega_t$ uses the exact frequencies of the current potential, making the $b = 0$ contours energy surfaces. The sine factor enforces the regularity condition $g \to 2$ as $L \to 0$ at any anisotropy amplitude (Binney, 2026). $b > 0$ favours radial orbits and $b < 0$ tangential orbits. $b(q)$ is an orbital-weight prescription, not $\beta(r)$.

The **one-transition** model (the original baseline) has

$$b(q) = b_{\mathrm{out}}\, S(q; J_a), \qquad S(q; J_t) = \frac{q^2}{q^2 + J_t^2}, \tag{4}$$

so $b \to 0$ for $q \ll J_a$ and $b \to b_{\rm out}$ for $q \gg J_a$. The **two-transition** model is

$$b(q) = b_{\rm out}\, S(q; J_a) + (b_{\rm outer} - b_{\rm out})\, S(q; J_{\rm outer}), \qquad J_{\rm outer} > J_a, \tag{5}$$

with limits 0, $b_{\rm out}$ and $b_{\rm outer}$. Both forms were defined in the model proposal; eq. (5) was implemented and unit-tested before this work. Stars carry both mass and light (mass follows light). There is one stellar population.

3.2 What the DF family can represent

Each parameter set defines one spherical, self-gravitating equilibrium. From it the model predicts, with no further assumptions:

- the intrinsic stellar density $\rho_\star(r)$ and its projection $\Sigma(R)$, compared with the photometry through one zero point (constant M/L);
- the intrinsic radial and tangential dispersions $\sigma_r(r)$ and $\sigma_t(r)$ (one component), hence $\beta(r) = 1 - \sigma_t^2/\sigma_r^2$;
- the projected LOS dispersion $\sigma_{\rm los}(R)$ and the two plane-of-sky dispersions $\sigma_{{\rm pm},R}(R)$ and $\sigma_{{\rm pm},T}(R)$. Their ratio $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$ is the most direct projected signature of anisotropy;
- the enclosed mass of every component and the full phase-space density $f(J_r, L)$.

It predicts no mean rotation (the DF is even in L_z), no flattening and no population differences.

Figure 1 shows what shapes the family can take. Starting from the two-transition best fit (Section 5.3), one ingredient is varied per row and each equilibrium is rebuilt self-consistently in the fitted remnant potential:

- **Envelope exponent** α (row a) sets how steeply the DF, and hence the density, falls with action. At fixed J_0 and $M_\star$, moving from $\alpha = 0.7$ to 1.4 shrinks the half-mass radius from 22 to 2 pc and raises the central LOS dispersion from 15 to 40 $\mathrm{km\,s^{-1}}$. It also changes the anisotropy the same $b(q)$ produces, because the orbits populate a different range of q.
- **Action scale** J_0 (row b) mainly sets the size. The half-mass radius is 5, 10 and 24 pc for $J_0 = 40$, 65 and 120 $\mathrm{pc\,km\,s^{-1}}$. At fixed mass, the central dispersion falls as the cluster grows.
- **One anisotropy transition** (row c) gives only monotonic profiles. $b_{\rm out} > 0$ makes β rise towards radial outside J_a, so $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$ falls below one. $b_{\rm out} < 0$ gives tangential outskirts, with the ratio rising above one. A single transition cannot turn from radial to tangential. The anisotropy is not independent of the light: at fixed J_0 and α the half-mass radius changes from 6.7 to 18 pc between $b_{\rm out} = -1$ and $+1$. The photometry and kinematics therefore have to be fitted jointly.
- **The second transition** (row d) adds the non-monotonic shape the data require (Section 5.2): a mild radial bump followed by tangential outskirts, seen in projection as a dip and then a rise of $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$. $J_{\rm outer}$ sets where the turnover occurs. At the best fit ($J_{\rm outer} = 812$ $\mathrm{pc\,km\,s^{-1}}$), β peaks at $+0.16$ near 9 pc and crosses zero at 23 pc. With $J_{\rm outer} = 300$ $\mathrm{pc\,km\,s^{-1}}$ the peak is only $+0.04$ (at 2 pc) and the zero crossing moves in to 4 pc.

A built-in property: every model is isotropic at the centre. $b(q) \to 0$ as $q \to 0$, and the regularity condition forces $g \to g_H$ there, so $\beta(0) = 0$ for every parameter choice. This follows from requiring the DF to be smooth at the phase-space origin (Binney, 2026). In a cored potential, a smooth DF is necessarily isotropic at $r = 0$. A tangentially biased core needs a DF that vanishes as $L \to 0$, and a radially biased core needs one that diverges. The other built-in properties are positivity, smoothness in action space and finite total mass.

This is an *assumption*; these data do not test it. β is an intrinsic, three-dimensional quantity. The data provide three projected dispersions at each projected radius R, each an integral along the line of sight over all $r \geq R$, so $\beta(r)$ can only be recovered through a dynamical model. In particular, the projected ratio $\sigma_{\mathrm{pm},T}/\sigma_{\mathrm{pm},R}$ tends to one as $R \to 0$ for *any* intrinsic anisotropy. The observed central value (about 1 inside 1 pc) therefore carries no information on $\beta(0)$. The central anisotropy enters mainly through the radial run of the dispersions and the ratio of LOS to PM dispersions. That ratio is degenerate with the distance, because converting a PM to $\mathrm{km\,s^{-1}}$ multiplies it by D.

The Jeans fits do not measure the central anisotropy either. Their central value, $\beta_0 = -0.5$, is the upper bound of the prior. It was imposed because a cored tracer around a central point mass requires $\beta_0 \leq -1/2$ for a positive DF (An and Evans, 2006). Their maximum-likelihood value, -0.503, lies on that bound. Neither model therefore provides a data-driven central anisotropy, and a test of the $\beta(0) = 0$ assumption is still to be done.

The property would matter in two cases. First, with a central black hole: around a point mass the regularity analysis changes (the derivative ratio becomes 1, not 2), and the cusp-slope–anisotropy theorem then requires either a tangential core or a stellar density cusp. Adding a BH therefore requires re-deriving the regular DF for a Kepler centre, not only adding a point-mass potential. Second, if a model-based analysis showed that the data prefer an anisotropic centre.

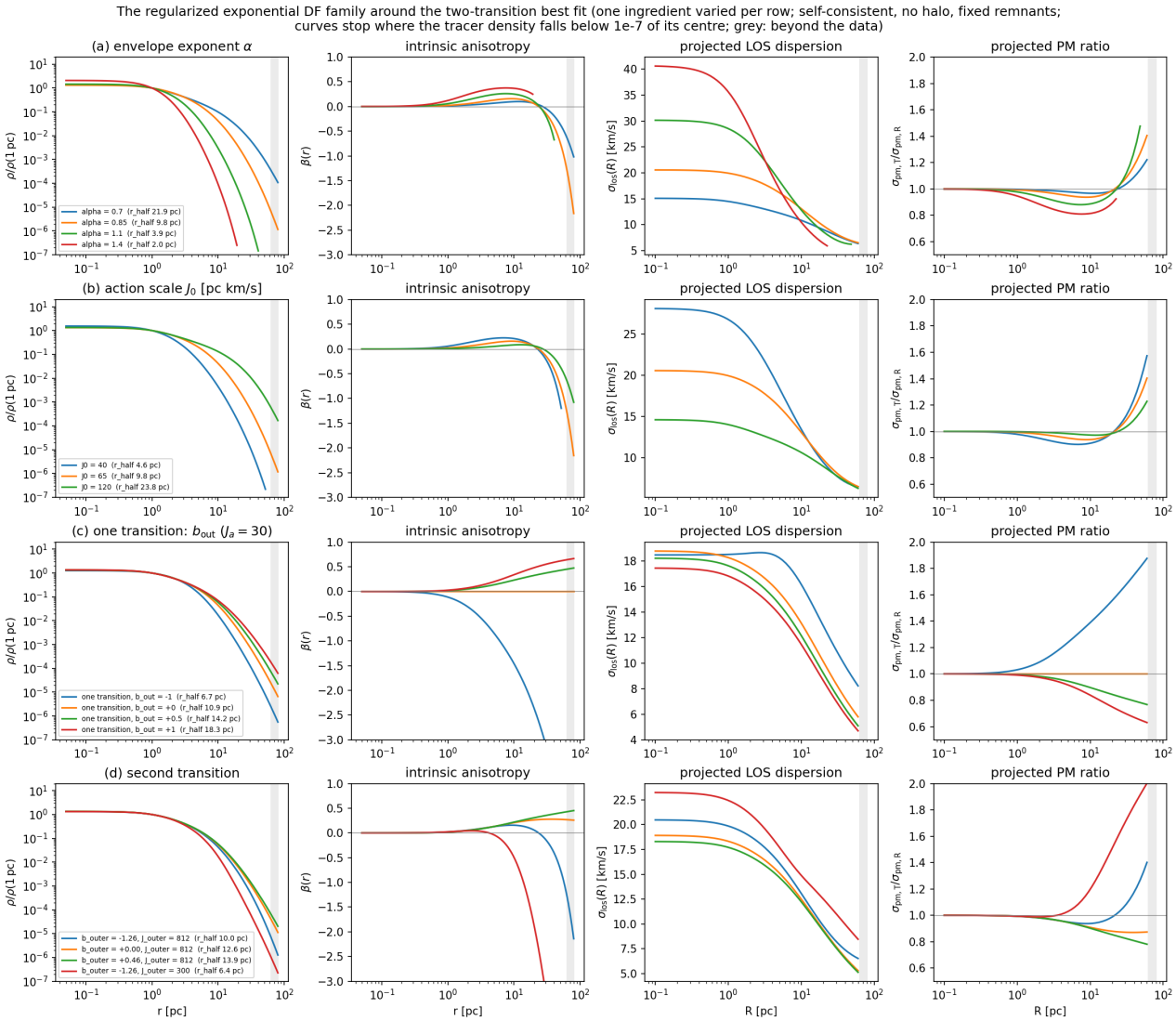


Figure 1: The regularized exponential DF family around the two-transition best fit. Rows vary (a) α, (b) J_0, (c) a single anisotropy transition with amplitude $b_{\rm out}$ at $J_a = 30\,\mathrm{pc\,km\,s^{-1}}$, and (d) the second transition ($b_{\rm outer}$, $J_{\rm outer}$). All other parameters stay at the best-fit values. Columns: intrinsic density normalized at 1 pc (legend: half-mass radius); intrinsic $\beta(r)$; projected $\sigma_{\rm los}(R)$; projected $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$. Every model is a self-consistent equilibrium in the fitted remnant potential (no halo, no point mass). Curves stop where the tracer density falls below 10^{-7} of its central value. Grey marks radii beyond the data. These are illustrations, not fits (script `bin/plot_df_family_gallery.py`).

3.3 Dark components

Dark mass enters as prescribed spherical densities in the same potential:

- **Remnants**: a Plummer sphere with mass $M_{\rm rem}$ and scale $a_{\rm rem}$.
- **Halo** (free-halo branch only): a cored profile ($\gamma = 0$) normalized at 20 pc, $\rho_{\rm DM}(r) = \rho_{20}\, h(r/r_s)T(r/r_t)/[h(20\,{\rm pc}/r_s)T(20\,{\rm pc}/r_t)]$, with $h(x) = (1+x)^{-3}$ and $T(y) = e^{-y^2}$. The scale r_s is free and the taper $r_t = 1000$ pc is fixed. The no-halo branch sets $\rho_{20} = 0$.
- **No central point mass** (Section 5.7).

3.4 Equilibrium, projection and numerics

For every trial, the stellar DF is rebuilt and iterated with Poisson's equation, with the dark densities held fixed, until the stellar density closes (AGAMA: Vasiliev, 2019). Projected surface density and LOS/PM second moments follow from the intrinsic moments, averaged over each bin's selection nodes. The fitting resolution uses 128 potential nodes, 32 velocity nodes, 240 moment nodes, 128 projection nodes and 192 action-contour nodes. The equilibrium iteration tolerance is 5×10^{-5} (Section 4.2). Every selected solution is validated at doubled resolution.

4 Fitting method and validation

4.1 Optimizer

Fits use bounded trust-region least squares (SciPy `least_squares`, TRF) on the residual vector of eq. (1) plus photometry. All coordinates are scaled to $[0, 1]$; mass and action scales are logarithmic. Jacobians are forward differences with step 0.002, each probe seeded with the base point's converged stellar potential. The 9–11 probes of a Jacobian run in parallel worker processes. A fit stops when the objective improves by less than 0.01 over two accepted steps (well below a meaningful $\Delta\chi^2 \sim 1$), when SciPy's own tests are met, or at 600 equilibrium evaluations. Stops that follow closely after rejected equilibria are recorded as unconverged. Several starting points are used for each model: two hand-picked and, for the wide-bound and two-transition runs, four Latin-hypercube starts (seed 20260923) in the central 60% of each range.

The second transition is parametrized by $\Delta = \ln(J_{\rm outer}/J_a) \in [\ln 2, \ln 100]$, applied after all other coordinates, so $J_{\rm outer} > J_a$ holds for every trial. The search bounds are listed in Table 2. They are optimizer limits, not priors.

Table 2: Search bounds. The "default" bounds were used in the first batch only.

Coordinate	Bounds	Notes
$M_\star$	10^6–$6 \times 10^6\,{\rm M}_\odot$ (log)	
J_0	30–800 pc km s^{-1} (log)	default 100–800
α	0.6–3	
$b_{\rm out}$	-2 to $+2$	± 1 in one-transition runs
J_a	5–500 pc km s^{-1} (log)	default 50–3000; 5–3000 in wide one-transition runs
$b_{\rm outer}$	-4 to $+2$	two-transition only
$\ln(J_{\rm outer}/J_a)$	$\ln 2$–$\ln 100$	two-transition only
$M_{\rm rem}$, $a_{\rm rem}$	10^2–$10^6\,{\rm M}_\odot$, 0.3–20 pc (log)	
ρ_{20}, r_s	0–10 ${\rm M}_\odot$ pc^{-3}, 5–500 pc (log)	free-halo only; default $r_s \le 150$

4.2 Numerical checks

- **Evaluation noise.** At a converged model, repeating the equilibrium with different seeds shifted residuals by up to 0.0060 errors at the original tolerance (5×10^{-4}), and Jacobian columns by 0.1–7%. At 5×10^{-5} these fall to 0.00059 errors and 0.01–0.65%, at 23% higher cost per evaluation. The tighter tolerance is used throughout.
- **Parallel Jacobians** reproduce serial ones: all probe objectives agree to 2.4×10^{-9} relative. The potential is transferred by file, a round trip exact to 3×10^{-14}. Probe wall time drops 4.6-fold.
- **Validation of each selected fit**: a cold replay, a rebuild at doubled resolution, and an independent direct projection over the data radii (0.05–68 pc). The thresholds are shifts below 0.1 errors and projection agreement within 0.5%. All solutions reported here pass. The largest values are a cold shift of 0.0046, a refinement shift of 0.012 errors, and a projection difference of 0.11%.

4.3 Acceptance criteria

The formal errors are known to be incomplete (Section 2.2), so structural and statistical success are judged separately.

- **Structural pass:** independent starts converge; no coordinate at a search bound; no coherent residual pattern; no dataset with $\chi^2/n > 3$; a physically smooth DF.
- **Statistical pass:** $\chi^2_{\rm kin}/N < 1.3$, every dataset $\chi^2/n < 2$, $\chi^2_{\rm phot}/N < 2$. This is required only under a justified error model; with the published errors it is reported, not required.

5 Results

5.1 The one-transition DF fails

Table 3 summarizes the one-transition fits. In every branch, all valid starts converge to the same minimum; five of five do so in each wide-bound branch. The kinematic fit remains poor. The anisotropy scale J_a sits at its lower bound in every fit, whether that bound is 50 or 5 pc km s^{-1}. Widening the bounds changes $\chi^2_{\rm kin}$ by only 29 (free halo) and 20 (no halo). Figure 2 shows the residual pattern. The tangential PM dispersion is too high at 3–7 pc (to −6 errors) and too low beyond 12 pc (to +6). The LOS dispersion is 3–5 errors too high at 5–8 pc, and the radial PM 2–5 errors too high beyond 20 pc.

Table 3: One-transition fits. The objective is $\chi^2_{\rm kin} + \chi^2_{\rm phot}$ after refinement. Dataset columns give χ^2/n.

Fit	Obj.	$\chi^2_{\rm kin}$	$\chi^2_{\rm kin}/N$	HST R	HST T	MUSE	Gaia R	Gaia T	At bound
Default, free halo	653.6	456.1	5.12	2.5	5.6	2.7	8.2	15.1	J_0, J_a, r_s
Default, no halo	717.6	475.7	5.34	2.3	6.8	2.8	2.6	20.1	J_a
Wide, free halo	623.3	427.5	4.80	2.1	4.5	2.6	10.1	13.6	J_a
Wide, no halo	698.1	456.1	5.12	2.1	6.2	2.7	2.6	20.2	J_a

5.2 Why it fails: diagnostic fits

Diagnostic fits of the same no-halo DF (Table 4) isolate the cause.

1. **The DF fits each half of the data.** HST+MUSE and Gaia are each fitted well by the same family. Only their combination fails.

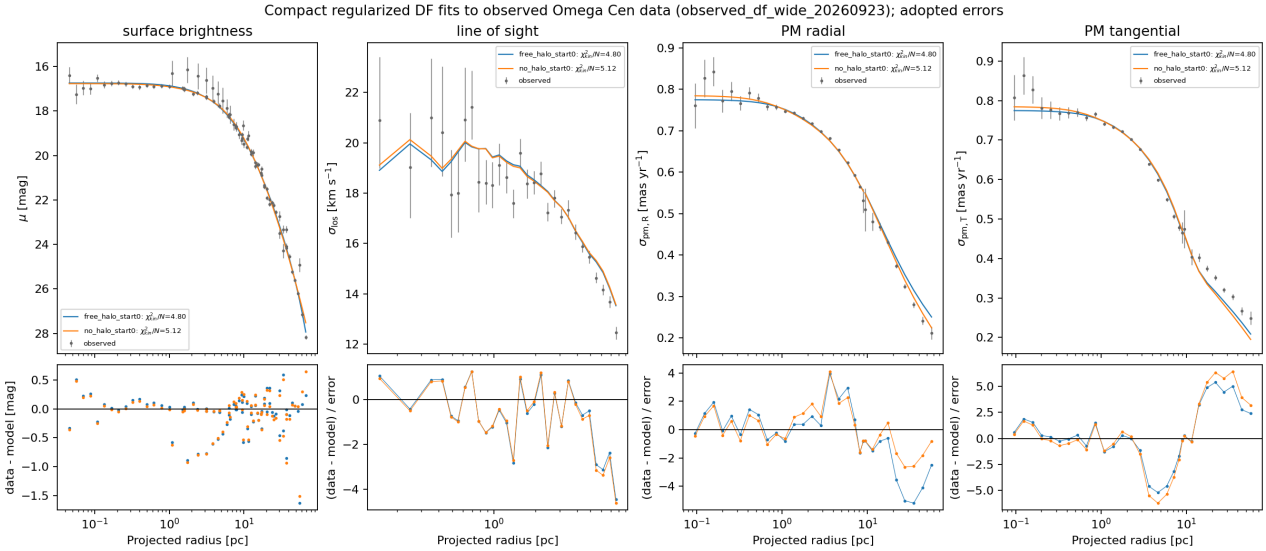


Figure 2: Best one-transition fits (wide bounds) with and without a halo. Top: surface brightness, LOS and PM dispersions with published errors. Bottom: normalized residuals (data−model)/error.

Table 4: Diagnostic fits of the one-transition, no-halo DF. Both starts of every fit converged to the same solution. The Jeans row is the rung-2 turnover fit (with a point mass, no DM) on the same 89 bins. (b): at a search bound.

Fit	$\chi^2_{\rm kin}/N$	HST R	HST T	MUSE	Gaia R	Gaia T	$b_{\rm out}$	J_a
All data	5.12	2.1	6.2	2.7	2.6	20.2	+0.18	5 (b)
HST+MUSE only (71 bins)	**1.43**	1.0	1.1	2.0	–	–	+0.39	32
Gaia only (18 bins)	**0.88**	–	–	–	0.8	1.0	−1.00 (b)	1665
All data, distance free	4.78	1.8	6.6	1.6	2.6	19.6	+0.18	5 (b)
Jeans turnover (ref.)	*2.13*	*2.9*	*1.7*	*1.8*	*1.9*	*2.6*		

2. **The halves demand opposite anisotropy.** The inner data want radial bias switching on at small actions ($b_{\rm out} = +0.39$, $J_a = 32\,{\rm pc\,km\,s^{-1}}$). The outer data want strongly tangential bias ($b_{\rm out} = -1$). One transition cannot supply both.

3. **Distance is not the cause.** Freeing D gives $D = 5.30$ kpc and lowers $\chi^2_{\rm kin}$ by only 31. Changing D rescales both PM components equally and cannot change σ_T/σ_R.

4. **An independent Jeans fit needs the same shape.** The rung-2 Jeans turnover fit prescribes $\beta(r) = 1 - \sigma_t^2/\sigma_r^2$ with two transitions. It finds $\beta \simeq +0.15$ near 6 pc, zero near 20 pc, and -0.6 to -0.7 at 80 pc. Its central value, $\beta_0 = -0.50$, sits at its prior limit (Figure 3).

The tangential turn lies at ~ 20 pc, inside the Gaia coverage, not at the HST/Gaia boundary.

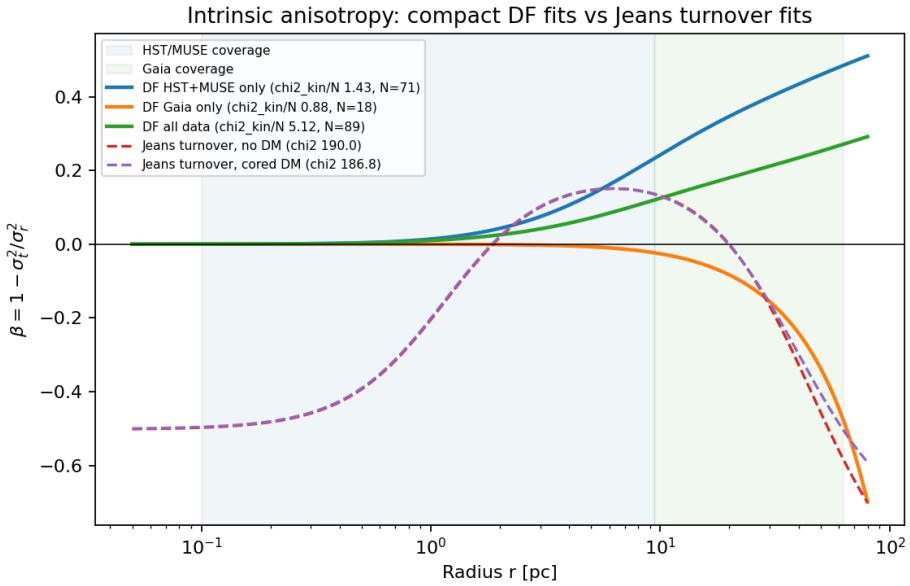


Figure 3: Intrinsic anisotropy $\beta(r)$ of the one-transition DF fitted to HST+MUSE only, to Gaia only, and to all data, compared with the rung-2 Jeans turnover fits. Shading marks the radial coverage of the two surveys.

Action coverage. The anisotropy acts on $q = J_r + L$. Sampling 2×10^5 tracers from the best all-data DF shows that the data sample $q \simeq 20$–$730\,{\rm pc\,km\,s^{-1}}$ (5–95%; Figure 4). The Gaia-only $J_a = 1665$ lies above even the outermost bin's 95th percentile. That fit therefore used only the low-q tail, $b \simeq b_{\rm out}(q/J_a)^2$, where only $b_{\rm out}/J_a^2$ is constrained. Neighbouring radial ranges overlap strongly in q, so a transition in q is soft in radius. These results set the bounds for the second transition (Table 2).

5.3 The two-transition DF without a halo

Six starts were run: the two hand-picked starts combine the two subset solutions, and four are Latin-hypercube starts. Two Latin starts were infeasible at their initial point. The four valid starts converged to one minimum (refined objective 378.3–378.4, $\Delta\chi^2_{\rm kin} \leq 0.6$), with no coordinate at a bound and no rejected evaluation. The best-fit parameters are $M_\star = 3.25 \times 10^6\,{\rm M_\odot}$, $J_0 = 65\,{\rm pc\,km\,s^{-1}}$, $\alpha = 0.85$, $b_{\rm out} = +0.46$, $J_a = 27\,{\rm pc\,km\,s^{-1}}$, $b_{\rm outer} = -1.26$, $J_{\rm outer} = 812\,{\rm pc\,km\,s^{-1}}$, $M_{\rm rem} = 1.72 \times 10^5\,{\rm M_\odot}$ and $a_{\rm rem} = 1.27$ pc.

$\chi^2_{\rm kin}$ falls from 456.1 to **159.2** ($\chi^2_{\rm kin}/N = 1.79$). Gaia tangential improves from $\chi^2/n = 20.2$ to 3.7 (Table 5). The intrinsic $\beta(r)$ follows the Jeans turnover to ~ 30 pc (Figure 5): $\beta = 0.02$, 0.14, 0.16, 0.06 and -0.13 at 1, 6, 10, 20 and 30 pc. Beyond ~ 40 pc it falls faster, reaching -1.27 at 63 pc. That region is driven by the $b_{\rm outer}$ asymptote, which acts only beyond the sampled q, so it is an extrapolation. The bias $b(q)$ peaks at $+0.41$ at $q = 103\,{\rm pc\,km\,s^{-1}}$, where the HST/MUSE tracers

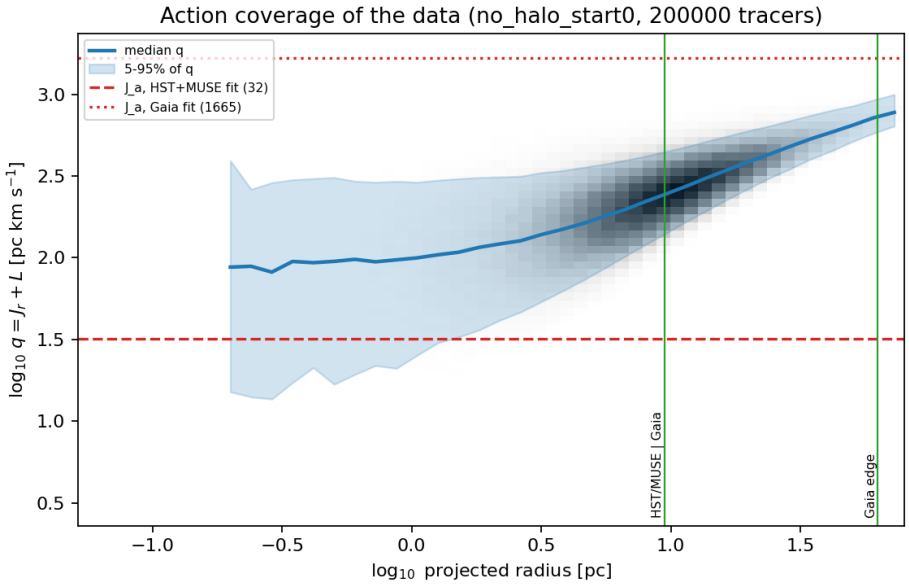


Figure 4: Distribution of $q = J_r + L$ of DF tracers against projected radius, with 5–95% quantiles. Horizontal lines mark the J_a of the HST+MUSE-only (32) and Gaia-only (1665) fits.

lie (q 5/50/95% = 56/170/379). It crosses zero at $q = 488$, inside the Gaia tracers (183/347/650; Figure 6). The DF is smooth and monotonic in the (J_r, L) plane, with no ridge at either transition. The remnant component returns to the inner-data values: the $6.6 \times 10^5\,\mathrm{M}_\odot$ remnant of the Gaia-only fit was compensating for the missing tangential outskirts.

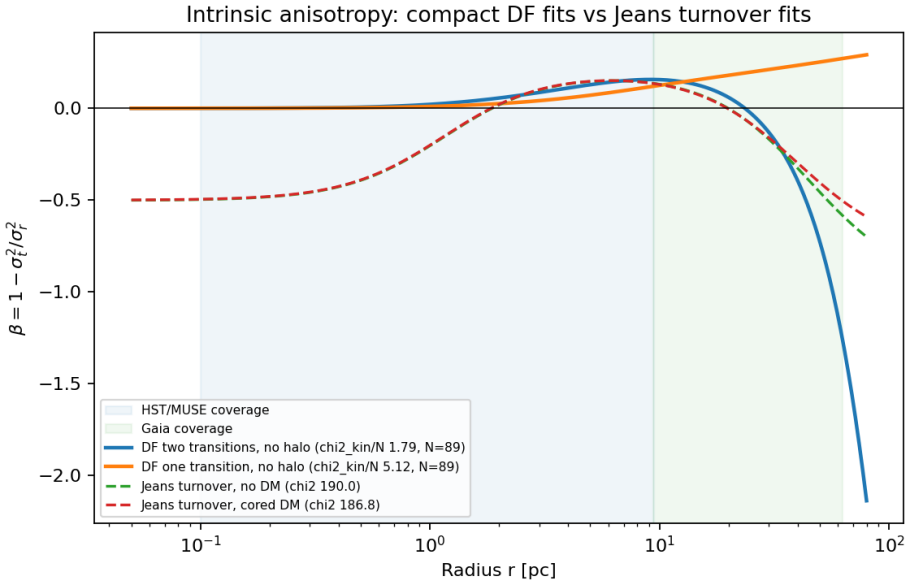


Figure 5: Intrinsic anisotropy of the two-transition fit (no halo) against the one-transition fit and the Jeans turnover fits.

5.4 Adding a free cored halo

With a free halo, three of six starts were infeasible at their initial point: two numerically (an action-normalization error of 5.1×10^{-5} above the 2×10^{-5} tolerance, and a density-closure failure) and one physically (the streaming term exceeded the DF second moment). The three valid starts converged to one minimum. The refined objective is 346.9, against 378.3 without a halo, and $\chi^2_{\mathrm{kin}} = 143.3$. The fitted parameters are $\rho_{20} = 0.92\,\mathrm{M}_\odot\,\mathrm{pc}^{-3}$, $M_\star = 3.21 \times 10^6\,\mathrm{M}_\odot$, $b_{\mathrm{out}} = 1.76$, $J_a = 101\,\mathrm{pc\,km\,s^{-1}}$, $b_{\mathrm{outer}} = -0.25$ and $J_{\mathrm{outer}} = 203\,\mathrm{pc\,km\,s^{-1}}$. However:

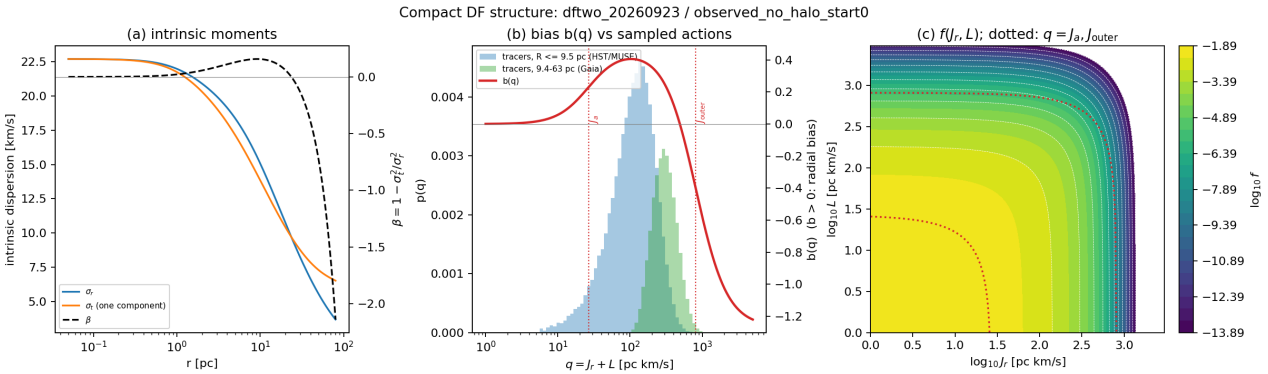


Figure 6: Structure of the two-transition fit (no halo). (a) Intrinsic σ_r, one-component σ_t and $\beta(r)$. (b) The bias $b(q)$ over the q distributions of tracers in the HST/MUSE and Gaia projected ranges. (c) $\log_{10} f$ in the (J_r, L) plane ($J_z = 0$); dotted lines mark $q = J_a$ and $q = J_{\rm outer}$.

- **Two coordinates are at bounds**: $r_s = 500\,\mathrm{pc}$ (upper), making the halo effectively uniform inside the data, and $J_{\rm outer}/J_a = 2$ (lower), pushing the two transitions together.
- **The anisotropy rearranges**: $b_{\rm out}$ changes from 0.46 to 1.76 and $b_{\rm outer}$ from -1.26 to -0.25.
- **The halo acts only in the outskirts**: it is under 1% of the enclosed mass at 20 pc, 8% at 42 pc and 32% at 80 pc. It trades misfit between the Gaia components: tangential improves from 3.7 to 2.3 while radial worsens from 0.9 to 2.0.

This is the classic mass–anisotropy trade. The objective gain (31, with $\Delta\chi^2_{\rm kin} = 16$ for two extra coordinates) cannot be read as a DM preference (Section 6).

5.5 Performance of the best-fit models

Table 5 and Figures 7–8 summarize the two best fits. The runs test counts sign changes of the residuals along radius; $z < -2$ means too few runs, i.e. coherent residuals.

Table 5: Performance of the two-transition fits with published errors. p is the χ^2 survival probability for n independent bins (indicative only). The last column gives the largest absolute normalized residual.

	no halo		free halo		runs z	max$\lvert r\rvert$
Dataset (n)	χ^2/n	p	χ^2/n	p	(no halo / halo)	(no halo / halo)
HST PM R (21)	1.01	0.45	1.02	0.43	-0.9 / 0.0	2.0 / 2.0
HST PM T (21)	1.81	0.013	1.39	0.11	-1.1 / -1.1	2.6 / 2.1
MUSE LOS (29)	2.01	0.001	1.84	0.004	+0.5 / +0.5	3.6 / 3.2
Gaia PM R (9)	0.93	0.50	2.04	0.03	-2.3 / -1.2	1.9 / 2.2
Gaia PM T (9)	3.70	10^{-4}	2.34	0.01	-0.8 / -1.0	3.6 / 2.7
Photometry (82)	2.67	2×10^{-14}	2.48	3×10^{-12}	-1.2 / -2.1	5.7 / 5.8
$\chi^2_{\rm kin}$ (89)		159.2		143.3		
$\chi^2_{\rm phot}$ (82)		219.1		203.6		

- **The kinematics are now fitted better than by the best Jeans model** ($\chi^2_{\rm kin} = 159.2$ and 143.3, against 190.0), with a positive, self-consistent DF. All profiles are tracked over 0.1–63 pc. The residual distribution is somewhat wider than $N(0,1)$, with tails to ±3.6.
- **The remaining structured misfits are localized.** The MUSE LOS is 2–3.6 errors too high at its outer edge (5–8 pc), where the streaming correction is largest. The tangential PMs are 2–2.6 errors too high at 4–7 pc and up to 3.5 errors too low at 15–20 pc (the first Gaia bins). With the halo, the outer Gaia radial PM becomes ~ 2 errors too high.

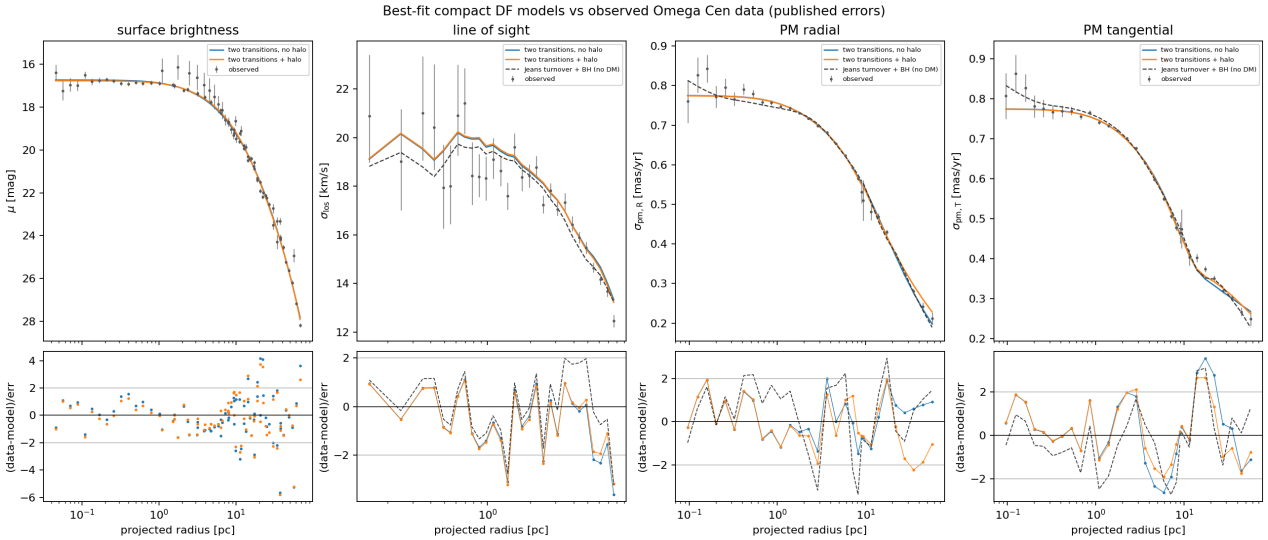


Figure 7: Best two-transition fits without (blue) and with (orange) a halo, and the rung-2 Jeans turnover fit (dashed; kinematics only). Bottom: (data−model)/error, with ±2 lines.

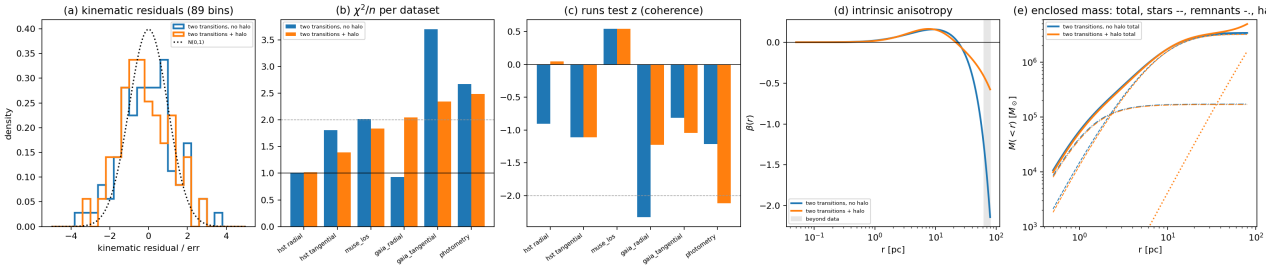


Figure 8: Diagnostics of the two best fits: (a) kinematic residuals against $N(0,1)$; (b) χ^2/n per dataset; (c) runs-test z per dataset; (d) intrinsic $\beta(r)$; (e) enclosed mass by component (total, stars, remnants, halo).

- **The photometric misfit** is dominated by 10–40 pc points with residuals of opposite sign at similar radii (-3 to -6 and $+2$ to $+4$). This is consistent with the internal scatter of the compilation (Section 2.2); this interpretation has not been tested.
- **The two models are nearly identical inside the data**: $\beta(r)$ and the total $M(<r)$ coincide to ~ 25 pc.

Verdict on objective 1. The two-transition DF is a structural near-pass. The only structural criterion it misses is Gaia tangential at $\chi^2/n = 3.7$ (without a halo), against a limit of 3. It is not a statistical pass under the published errors: $\chi^2_{\rm kin}/N = 1.79$ and 1.61.

5.6 The two-transition DF with Poisson counts and the distance prior

The same two-transition DF was refitted with the count likelihood of Section 2.3 and the distance as a coordinate with a Gaussian prior 5.43 ± 0.05 kpc (Baumgardt and Vasiliev, 2021), entering the objective as $[(D - 5.43)/0.05]^2$. No halo. Four of six starts (two hand-picked, two Latin) converged to one minimum; two Latin starts were infeasible. The refined objective is 285.6: $\chi^2_{\rm kin} = 158.2$, HST count deviance 98.8 (20 bins), Gaia count deviance 26.3 (12 bins), prior 2.2. No coordinate is at a bound and the numerical checks pass.

Table 6: Counts-based fit (no halo). Kinematic rows: χ^2/n; count rows: Poisson deviance per bin; p as in Table 5.

Dataset (n)	value	p	runs z	max$\lvert r\rvert$
HST PM R (21)	2.07	0.003	-1.2	3.3
HST PM T (21)	1.95	0.006	-2.5	2.8
MUSE LOS (29)	1.31	0.12	-0.5	2.5
Gaia PM R (9)	1.55	0.12	-1.2	1.9
Gaia PM T (9)	2.42	0.010	$+0.4$	3.0
HST counts (20)	4.94	2×10^{-12}	-1.6	4.8
Gaia counts (12)	2.19	0.010	-0.6	2.9
$\chi^2_{\rm kin}$ (89)	158.2			$\chi^2_{\rm kin}/N = 1.78$

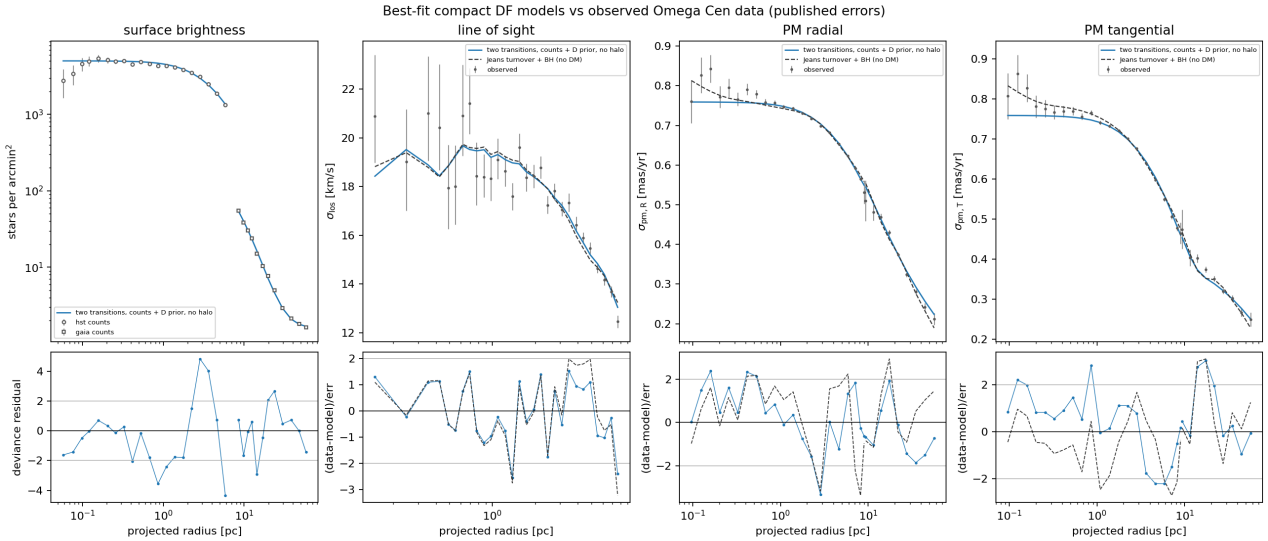


Figure 9: The counts-based two-transition fit (no halo). Left: HST and Gaia surface number densities with Poisson errors and the model μ_i/A_i; bottom row: deviance residuals for the counts, (data−model)/error for the kinematics. Dashed: the rung-2 Jeans turnover fit (kinematics only).

- **Kinematics**: quality unchanged ($\chi^2_{\rm kin}$ 158.2 vs 159.2) but more evenly spread: every dataset lies between 1.3 and 2.4 (Gaia tangential 2.4, from 3.7). All 89 residuals are within ± 3.3 errors.

- **Counts**: the Gaia profile is fitted at 2.2 per bin. The HST profile, with 118k stars and 0.3–1% Poisson errors, is the worst-fitted term: the model is $\sim 4\%$ low at 2–4 pc, $\sim 4\%$ high at 5–6.5 pc, and low inside 0.1 pc. This is the one-parameter envelope of Section 3.2 showing its rigidity at the few-per-cent level.
- **Parameters**: $M_\star = 3.00 \times 10^6\,\mathrm{M}_\odot$, $J_0 = 54\,\mathrm{pc\,km\,s^{-1}}$, $\alpha = 0.80$, $b_{\rm out} = +0.74$, $J_a = 64\,\mathrm{pc\,km\,s^{-1}}$, $b_{\rm outer} = -0.33$, $J_{\rm outer} = 319\,\mathrm{pc\,km\,s^{-1}}$, $M_{\rm rem} = 2.62 \times 10^5\,\mathrm{M}_\odot$ at 2.15 pc, $D = 5.356$ kpc (1.5σ below the prior).
- **The outer anisotropy changes.** Against the Trager-photometry fit ($b_{\rm outer} = -1.26$, $J_{\rm outer} = 812$), $\beta(r)$ agrees inside 30 pc but reaches only -0.45 at 80 pc instead of -2.1. The steep tangential outskirts of Section 5.3 were an artefact of the arbitrary photometric weighting. The parameter count is unchanged.

5.7 Is a central point mass needed?

The model has no central black hole. Figure 10 compares the 43 kinematic bins inside 2 pc with the no-BH two-transition DF and the rung-2 Jeans fits, which include a free point mass with maximum-likelihood $M_\bullet = 4.4 \times 10^4\,\mathrm{M}_\odot$ (16–84%: 3.9–4.6×10^4).

Table 7: χ^2 of the 43 kinematic bins inside 2 pc.

Model	total	HST R (13)	HST T (13)	MUSE (17)
DF, two transitions, no BH, no halo	**48.8**	11.7	12.5	24.6
Jeans turnover + BH, no DM	59.0	21.5	16.7	20.8
Jeans turnover + BH, cored DM	59.3	23.7	14.1	21.4

The binned data give no evidence that the DF needs a point mass. Only two HST bins (0.12–0.16 pc) lie 1–2 errors above the no-BH model, and the innermost bin (0.10 pc) lies below it. The comparison is not controlled: the Jeans fits also differ in anisotropy (with $\beta_0 = -0.5$ imposed) and in mass model. The clean test would be the same DF with and without a BH. The binned profiles also cannot test the published IMBH evidence. That evidence rests on individual fast-moving stars inside ~ 0.1 pc, which set a lower limit of $\sim 8200\,\mathrm{M}_\odot$ (Häberle et al., 2024), and those stars are not in these profiles. Adding a BH is therefore deferred.

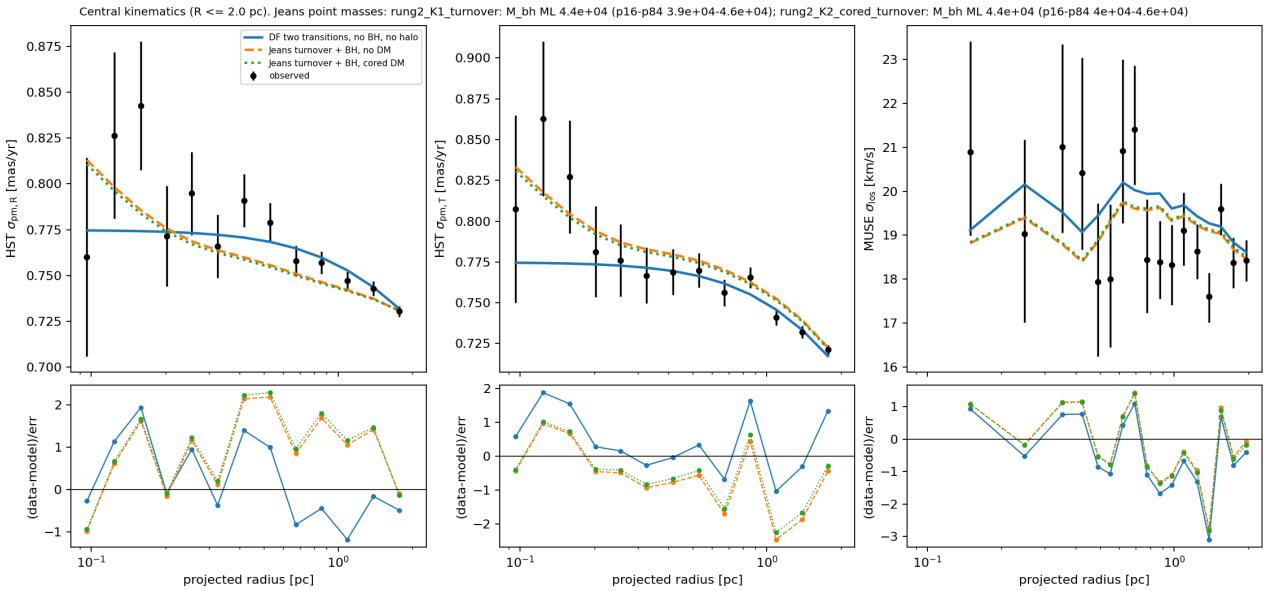


Figure 10: Central kinematics ($R \leq 2$ pc) with the no-BH two-transition DF and the Jeans turnover fits with a point mass. Bottom: (data−model)/error.

6 The dark-matter question: what can and cannot be said

All statements here use the counts-based likelihood of Section 5.6. Earlier halo fits (Section 5.4) used the arbitrary photometric weighting and are superseded.

6.1 Free halo

With ρ_{20} and r_s free, seeded from the no-halo best fit, the optimizer converges in 29 evaluations to $\rho_{20} = 0$ (at the bound), the no-halo solution, objective 285.6. The one Latin start was infeasible.

6.2 Profile likelihood in ρ_{20}

ρ_{20} was fixed at 0.25, 0.5, 1, 2 and 4 $\mathrm{M_\odot\,pc^{-3}}$ and every other coordinate, including r_s (5–500 pc), the anisotropy and the distance, refitted from the best fit plus one Latin start (Table 8, Figure 11).

Table 8: Profile likelihood in ρ_{20}. Δ is the refined objective minus its minimum. Halo masses are for the profiled halo. For $\rho_{20} = 0.25$–2 only the seeded start converged (the Latin start was infeasible); at 4 the Latin start improved the objective from 297.1 to 296.1.

ρ_{20}	Δ	χ^2_{kin}	counts dev.	r_s [pc]	$M_{\mathrm{DM}}(<20\,\mathrm{pc})$	$M_{\mathrm{DM}}(<63\,\mathrm{pc})$	$M_\star$
0 (free halo)	0	158.2	125.2	–	0	0	3.00×10^6
0.25	+0.51	158.9	125.0	5 (bound)	2.4×10^4	6.2×10^4	2.96×10^6
0.5	+1.05	159.3	125.1	5	4.8×10^4	1.2×10^5	2.92×10^6
1	+2.20	161.1	124.4	5	9.6×10^4	2.5×10^5	2.83×10^6
2	+4.78	164.4	123.6	5	1.9×10^5	4.9×10^5	2.67×10^6
4	+10.55	171.0	122.7	5	3.8×10^5	9.8×10^5	2.35×10^6

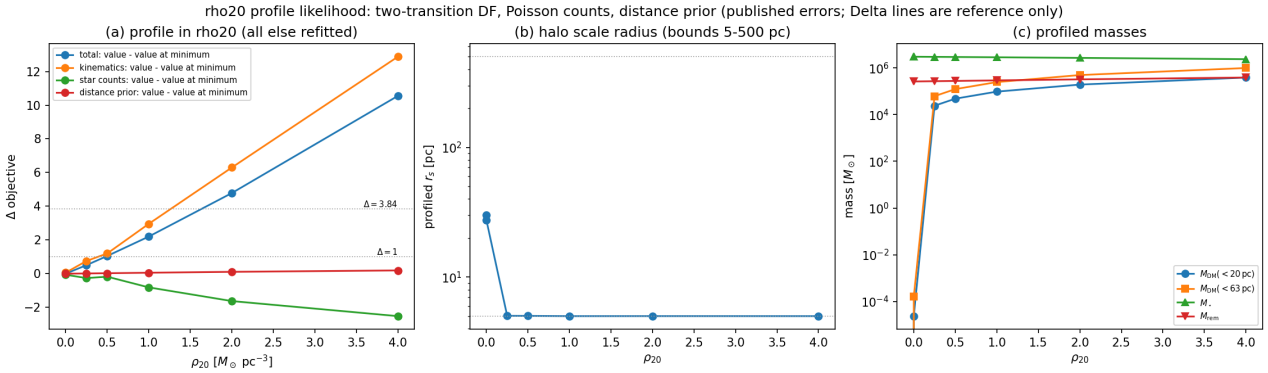


Figure 11: (a) The objective and its components relative to the minimum against ρ_{20}; the $\Delta = 1$ and 3.84 lines are reference values, not calibrated intervals. (b) The profiled r_s. (c) Profiled masses.

- **The data prefer no halo, monotonically.** The cost of a forced halo comes entirely from the kinematics; the count deviance falls slightly and the stellar mass drops from 3.0 to $2.35\times10^6\,\mathrm{M_\odot}$ across the scan: the star/DM trade, now quantified.
- **Reference values.** On a χ^2 scale, $\Delta = 1$ falls near $\rho_{20} \simeq 0.5$ and $\Delta = 3.84$ near $\rho_{20} \simeq 1.7$ ($M_{\mathrm{DM}}(<20\,\mathrm{pc}) \simeq 1.6\times10^5\,\mathrm{M_\odot}$). These would be 68% and 95% one-parameter limits only under a calibrated likelihood. $\chi^2_{\mathrm{kin}}/N = 1.78$ says the published errors are not, so the real limits are looser. No rescaling has been applied.
- **The halo is profiled to its least visible shape.** r_s runs to the 5 pc bound at every fixed ρ_{20}: with the density normalized at 20 pc, a compact halo has the least mass inside the data. The profile is therefore a constraint on the least harmful halo per ρ_{20}. A physically motivated extended remnant would be constrained more strongly; no such prior is imposed, because no clear expectation for r_s exists.

- **Comparison.** The earlier Jeans posterior gave a 95% upper limit $\rho_{20} = 1.03$ under different assumptions; the two methods agree on the order of magnitude.

6.3 Mock validation and coverage

Mocks were built from the counts-based best fit, each with the published kinematic noise (Gaussian per bin) and Poisson counts drawn with the fitted amplitudes. Truth A is the fitted model itself ($\rho_{20} = 0$); truth B adds a halo with $\rho_{20} = 1$, $r_s = 30\,\mathrm{pc}$. Six noise realizations of each were fitted with a free halo from generic hand-picked starts, never from the truth (Figure 12; `bin/run_mock_coverage.sh`).

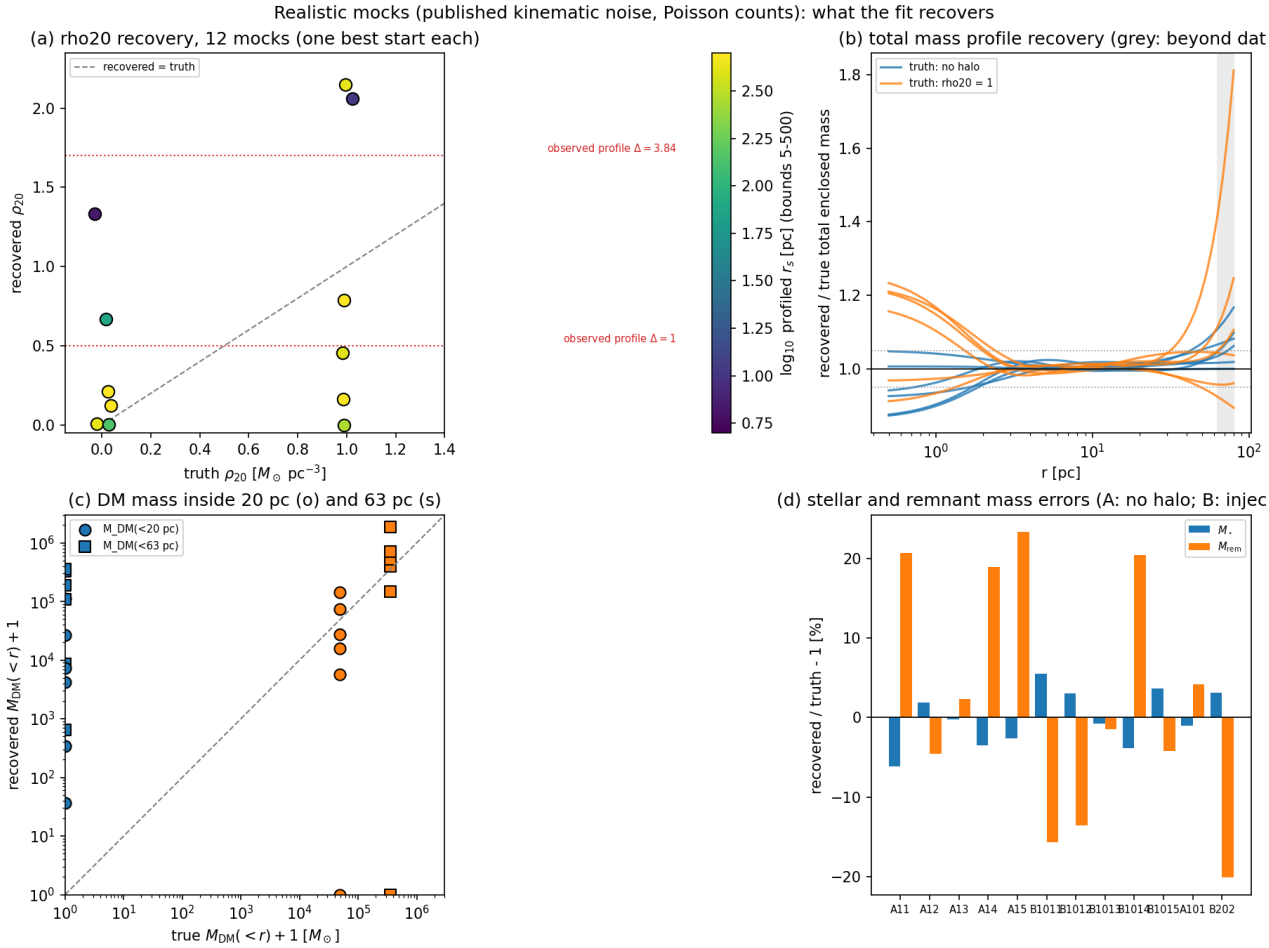


Figure 12: Recovery in twelve realistic mocks. (a) Recovered against true ρ_{20} (colour: profiled r_s); the red lines mark where the observed profile crosses $\Delta = 1$ and 3.84. (b) Recovered/true total enclosed mass against radius (grey: beyond the data). (c) Recovered against true DM mass inside 20 and 63 pc. (d) Fractional errors of the stellar and remnant masses.

- ρ_{20} **is weakly identified.** For truth A the recovered values are 0.00, 0.01, 0.12, 0.21, 0.67 and 1.33 (mean 0.39, scatter 0.52); for truth B they are 0.00, 0.16, 0.46, 0.79, 2.06 and 2.15 (mean 0.94, scatter 0.94). The estimator is roughly unbiased in the mean but its scatter is as large as the signal: one no-halo realization returned $\rho_{20} = 1.33$ (a compact 7 pc halo taking 6% of the stellar mass and adding 21% to the remnants), and one $\rho_{20} = 1$ realization returned exactly zero.

- **The total mass profile is what the data measure.** In all twelve mocks the total enclosed mass is recovered within ±3% at 20 pc and within ±5% over 2–40 pc, whatever the fitted split between stars, remnants and halo. Inside 1 pc and beyond 60 pc the recovery degrades (to ±25% and, in one case, +42% at 63 pc when a $\rho_{20} = 2$, $r_s = 420\,\mathrm{pc}$ halo was fitted).

- **The trade runs through the dark components and $M_\star$.** Stellar masses are recovered within ±6%, remnant masses within ±23%, and the errors are anti-correlated with the fitted halo.

Consequence for the observed profile (Section 6.2): its rise of $\Delta = 2.2$ at $\rho_{20} = 1$ lies within the estimator's noise scatter and cannot be quoted as a limit on ρ_{20}. What these data determine is the total mass profile over 2–40 pc; the stellar/DM split at 20 pc is not resolved with a free halo scale. Breaking the degeneracy needs information the kinematics do not carry: a shape separation between DM (extended, not mass-segregated) and remnants (compact), and an external prior on $M_\star$ from the stellar mass function (Section 9).

6.4 An external stellar mass: the present-day mass function

The degeneracy of Section 6.3 runs through the stellar mass. The HST count amplitude fitted in the dynamics, a, fixes how much mass that follows the light each counted star carries: $M/N_{19} = (\text{pc per arcsec})^2/a = 10.67\,\mathrm{M}_\odot$ per star with $F625W < 19$ (the count selection) for the counts-based best fit. We measured the same quantity from the oMEGACat photometry (`bin/build_mass_function.py`; Figure 13).

Method. MIST v1.2 isochrones (Dotter, 2016; Choi et al., 2016) evaluated with `minimint` (`https://github.com/segasai/minimint`), with the HST ACS/WFC $F625W$ and $F814W$ bolometric-correction tables prepared into a separate prefix; 12.5 Gyr, $D = 5.43$ kpc ($(m - M)_0 = 13.674$), $E(B-V) = 0.12$ applied through the tables' A_V axis ($A_{625} = 0.315$, $A_{814} = 0.219$). Unflagged stars (the quality flags are magnitude- and radius-dependent) in five zones 10–250 arcsec. The main-sequence ridge is matched best by [Fe/H] = −1.20 (colour rms 0.004 mag) against 0.009 mag for the Harris value −1.53; the two are degenerate with $E(B-V)$ at the 0.01 mag level and both are carried (turnoff masses 0.789 and 0.779 $\mathrm{M}_\odot$).

Completeness. With zone-dependent faint limits the slope flattened outwards, the opposite of mass segregation. With one common limit, $F625W < 21.0$ ($m > 0.52\,\mathrm{M}_\odot$, complete in every zone), the slope of $\mathrm{d}N/\mathrm{d}m \propto m^\alpha$ over 0.52–0.78 $\mathrm{M}_\odot$ steepens outwards monotonically: $\alpha = -1.37, -1.52, -1.56, -1.59, -1.77$ (±0.10 to ±0.03) from 10–30 to 175–250 arcsec at [Fe/H] = −1.20, and 0.13–0.14 shallower at −1.53.

Table 9: Mass that follows the light per star with $F625W < 19$, zone 60–100 arcsec, for the two isochrone metallicities. White dwarfs: IMF slope −2.3 above the turnoff, mean mass 0.55 $\mathrm{M}_\odot$ ($N_{\mathrm{WD}}/N_{19} = 4.6$–$4.7$). An IMF continuous with the fitted slope above the turnoff implies $N_{\mathrm{NS}}/N_{19} \simeq 1.4$ and $N_{\mathrm{BH}}/N_{19} \simeq 0.9$ and is rejected. NS/BH belong to the free remnant component and are not included.

Component [$\mathrm{M}_\odot$ per counted star]	[Fe/H] = −1.20	−1.53
Main sequence observed, 0.52–0.78 $\mathrm{M}_\odot$	1.80	1.77
Main sequence extrapolated to 0.1 $\mathrm{M}_\odot$: fitted slope	5.00	4.21
Kroupa (−1.3 below 0.5 $\mathrm{M}_\odot$)	4.22	3.92
flat (illustrative lower bound)	2.03	1.92
Evolved stars	0.11	0.10
White dwarfs	2.55	2.60
Total, fitted slope / Kroupa / flat	9.47 / 8.69 / 6.49	8.69 / 8.40 / 6.39

Across zones the fitted-slope total rises from 8.3 to 11.3 (7.7 to 10.3 at −1.53) from the inner to the outer zone. In the zones that dominate the kinematics (30–175 arcsec) the photometric value is 8.3–9.8, 12–27 per cent below the dynamical 10.67 (Figure 14). Two terms carry most of the budget and its uncertainty: the extrapolation below 0.52 $\mathrm{M}_\odot$ (3.4–6.6 per counted star) and the white dwarfs (2.5–2.7). A radially varying mass per bright star also means that a single mass-follows-light stellar DF cannot be exactly right.

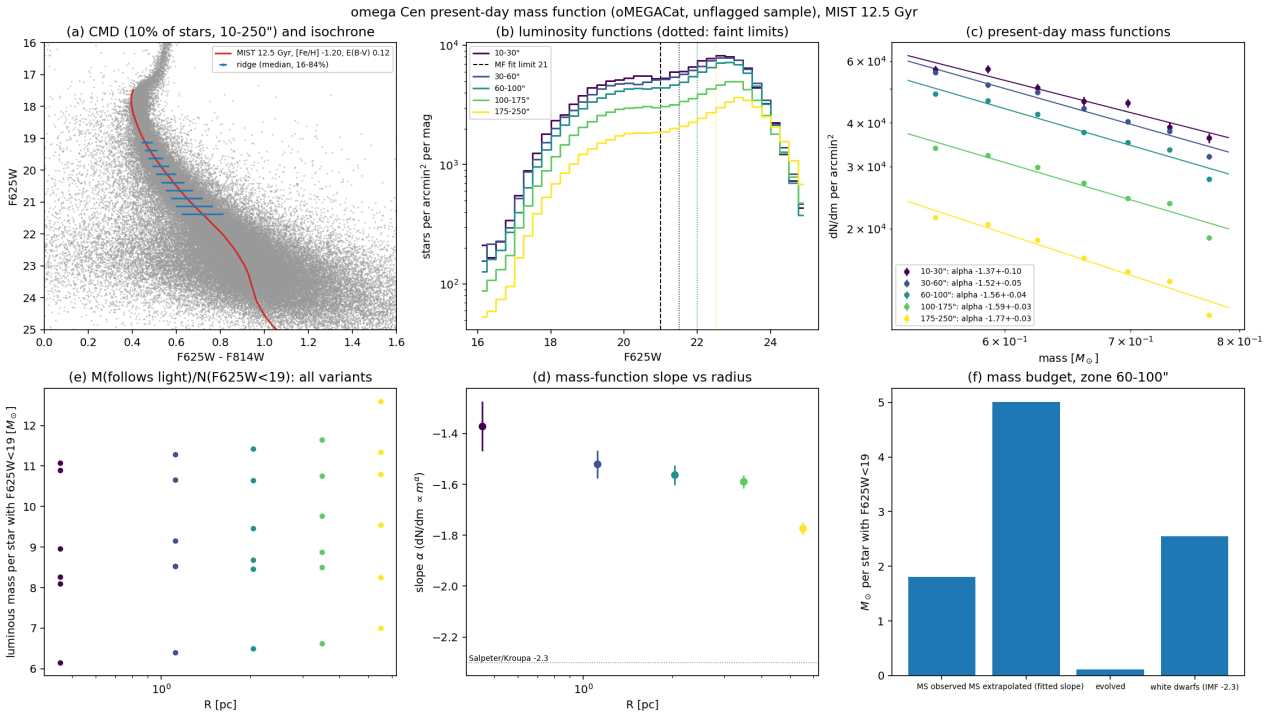


Figure 13: Present-day mass function from oMEGACat: colour–magnitude diagram with the adopted isochrone, luminosity and mass functions per zone, slope against radius (common faint limit $F625W < 21$), and the mass per counted star for the extrapolation and white-dwarf variants.

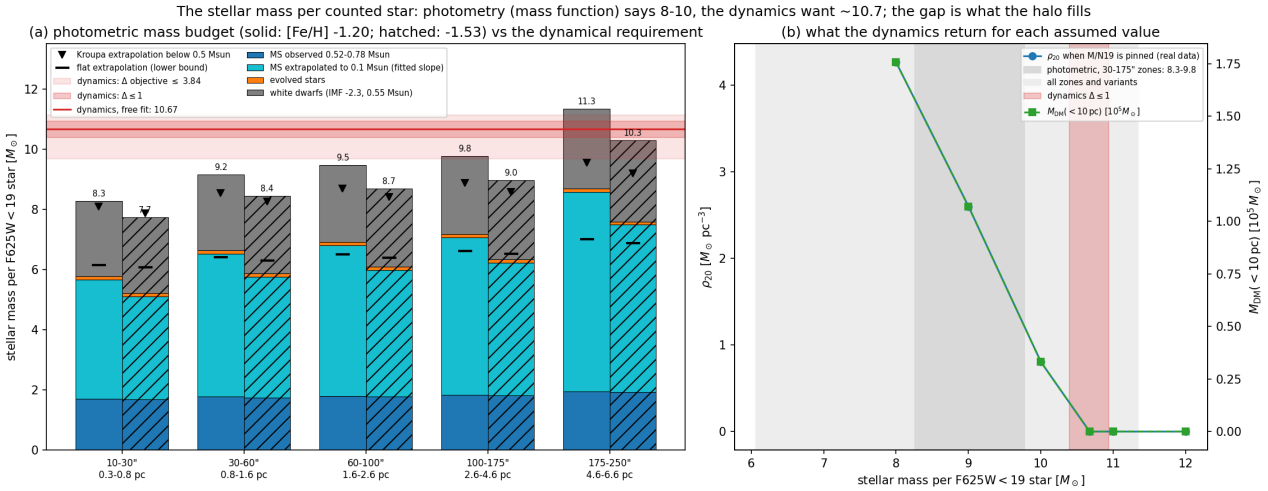


Figure 14: (a) Photometric mass per $F625W < 19$ star by component and zone (solid: [Fe/H] $= -1.20$; hatched: -1.53; markers: Kroupa and flat extrapolations) against the dynamical value (red line: free fit 10.67; bands: $\Delta \leq 1$ [10.39, 10.94] and $\Delta \leq 3.84$ [9.68, 11.14] from the scan of Section 6.5). (b) ρ_{20} and $M_{\rm DM}(< 10\,\mathrm{pc})$ returned by the real-data scan for each pinned value, with the photometric ranges shaded.

6.5 ρ_{20} as a function of the stellar mass per counted star

Rather than impose the photometric value through a Gaussian prior whose width is not yet defensible, the result is reported as a function of M/N_{19} (a decision of the project). The HST count amplitude is held at $a = (\text{pc per arcsec})^2/(M/N_{19})$ at the trial distance while the Gaia amplitude and field stay profiled; the HST counts then fix the stellar normalization and the kinematics must be met by the DF shape, the remnants and the halo (`-mass-per-star` in the driver). The scan is also the profile likelihood in M/N_{19}: adding a photometric term $((M - M_{\rm phot})/\sigma_{\rm phot})^2$ afterwards is equivalent to fitting with it.

Mocks. The coverage mocks were refitted with the amplitude pinned at the truth (10.67) and at wrong values (Figure 15). At the truth, $M_\star$ is recovered to < 1 per cent, but ρ_{20} still scatters: truth A returns 0.45, 0.006, 0.002, 0.15, 0.29 and truth B ($\rho_{20} = 1$) returns 1.39, 0.49, 1.89, 2.03, 0.53. A wrong value biases ρ_{20} strongly: $M/N_{19} = 9.0$ (-16 per cent) returns $\rho_{20} = 2.4$–3.1 for no-halo truths and 3.7–4.4 for $\rho_{20} = 1$; 12.0 ($+12$ per cent) returns $\rho_{20} \simeq 0$ for both. The slope is $\simeq -1.5\,\mathrm{M}_\odot\,\mathrm{pc}^{-3}$ per $\mathrm{M}_\odot$ per counted star, while $M(< 20\,\mathrm{pc})$ stays within $+6$ per cent in every case. For the external constraint to be decisive M/N_{19} must be known to $\sim 0.3\,\mathrm{M}_\odot$ (3 per cent).

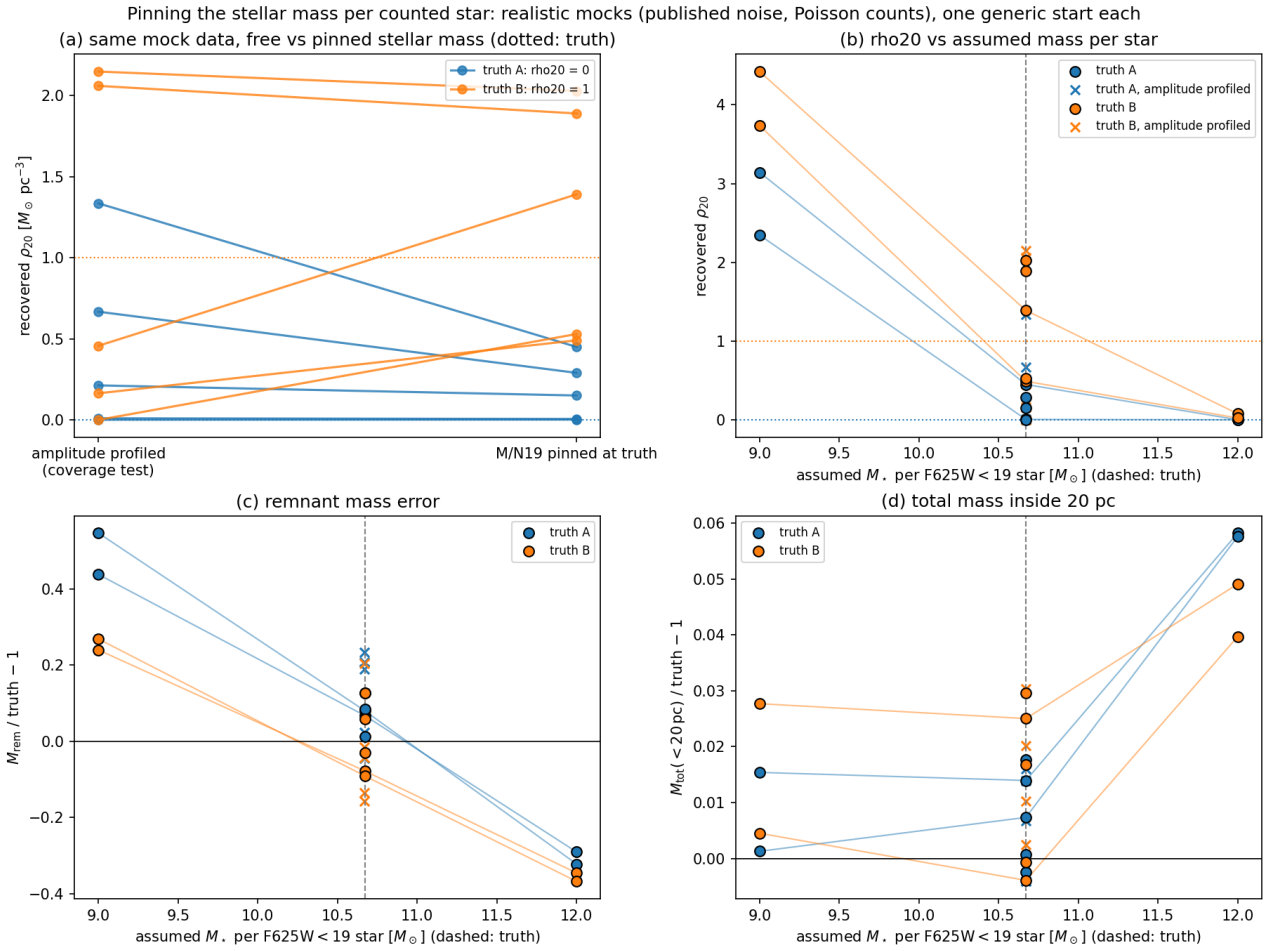


Figure 15: Mocks with the stellar mass per counted star pinned. (a) Same mock data, amplitude profiled (coverage test) against pinned at the truth. (b)–(d) Recovered ρ_{20}, remnant-mass error and $M(< 20\,\mathrm{pc})$ error against the assumed M/N_{19} (truth 10.67, dashed).

Observed data. Table 10 and Figure 16. The Latin starts were infeasible under a pinned amplitude, so each point rests on the seeded start; the fits at 11 and 12 were restarted once from their own best point after stopping early.

Table 10: Real-data scan in M/N_{19} (HST amplitude pinned, free cored halo, distance prior). Δ: objective minus the free-amplitude minimum. An asterisk marks r_s at its 5 pc bound.

M/N_{19}	Δ	$\chi^2_{\rm kin}$	ρ_{20}	r_s [pc]	$M_\star$	$M_{\rm rem}$ (a [pc])	$M_{\rm DM}(<10\,{\rm pc})$
8.00	+12.2	178.6	4.27	5*	2.24×10^6	4.37×10^5 (2.76)	1.76×10^5
9.00	+6.9	170.8	2.60	5*	2.52×10^6	3.68×10^5 (2.56)	1.07×10^5
10.00	+2.4	164.7	0.80	5*	2.82×10^6	3.09×10^5 (2.36)	3.3×10^4
10.67 (free)	0	158.2	0	–	3.00×10^6	2.62×10^5 (2.15)	0
11.00	+1.2	155.3	0	15.5	3.06×10^6	2.33×10^5 (2.00)	0
12.00	+19.7	158.6	0	78	3.22×10^6	1.64×10^5 (1.58)	0

Below the free value the halo switches on with $\mathrm{d}\rho_{20}/\mathrm{d}(M/N_{19}) \simeq -1.7$, r_s at its bound and the remnants growing with it, exactly as in the mocks; above it the halo stays off and the Gaia counts pay. Combined with a photometric term centred on 8.9 (the 30–175 arcsec value): for $\sigma_{\rm phot} = 0.5$ the minimum moves to 9.47 [8.98, 9.97] with $\rho_{20} = 1.8$ [0.9, 2.6] and $M_{\rm DM}(<10\,{\rm pc}) = 7\times10^4\,{\rm M}_\odot$; for $\sigma_{\rm phot} = 1.0$ and 1.5 it stays at 10.67 with $\rho_{20} = 0$ ([0, 1.2] and [0, 0.55]). The answer is therefore set by $\sigma_{\rm phot}$. At 0.5 the photometry produces a "detection" of the size a −16 per cent mass-per-star error produces on a no-halo mock; at ≥ 1 the data pull the minimum back to the free value. The next step is the photometric error budget, not more fitting.

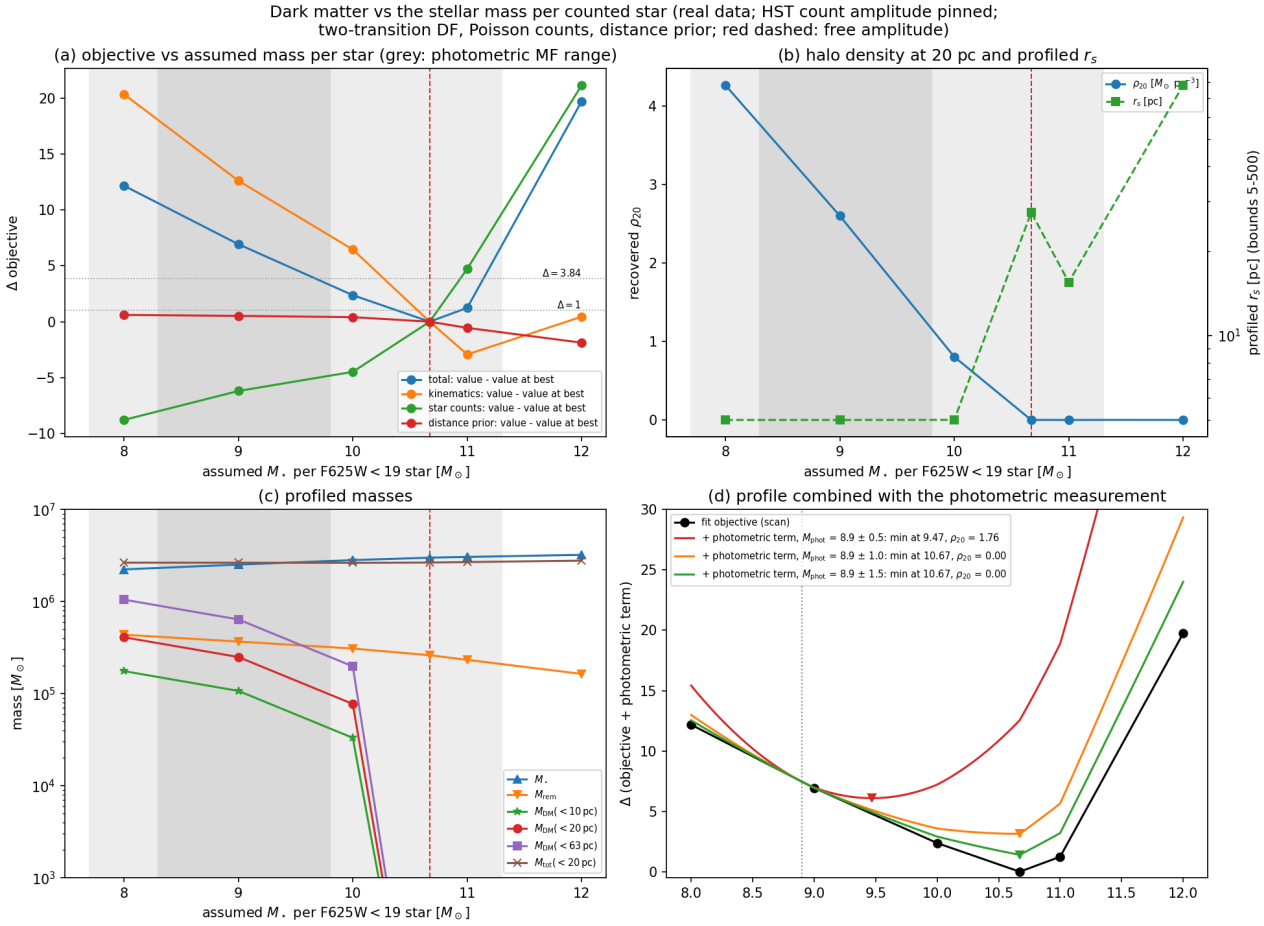


Figure 16: Real data against the assumed stellar mass per counted star. (a) Objective and components relative to the minimum (grey: photometric range). (b) ρ_{20} and the profiled r_s. (c) Profiled masses. (d) The scan combined with the photometric term for $\sigma_{\rm phot} = 0.5$, 1.0 and 1.5 (triangles: minima).

6.6 Anatomy of the degeneracy

Why a halo is invisible. The best no-DM fit and the refitted $\rho_{20} = 1$ ($r_s = 5\,\mathrm{pc}$) fit differ by at most 0.27σ in any bin of any dataset (Figure 19, left). Adding the same halo to the no-DM model *without* refitting changes the HST radial proper-motion dispersions by 2 per cent (12.5σ), the tangential ones by 6.5σ, and raises the objective from 283.4 to 1123.9; the refit brings it back to 285.5 by removing 5.5 per cent of the stellar mass, adding 11 per cent to the remnants and adjusting the anisotropy transitions (Figure 17). The coincident curves are the degeneracy itself.

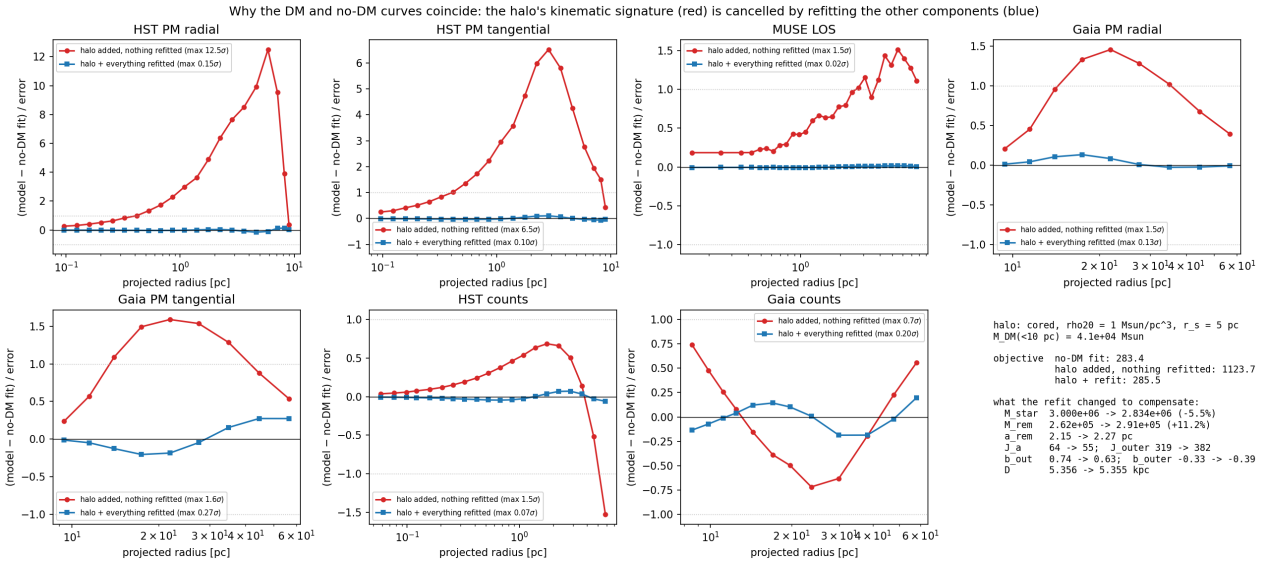


Figure 17: (Model − no-DM fit)/error per bin for a $\rho_{20} = 1$, $r_s = 5\,\mathrm{pc}$ halo added with nothing refitted (red) and with every other parameter refitted (blue). The text panel lists the compensating changes.

Who carries the mass inside 10 pc. Along the data scan in ρ_{20} the total inside 10 pc changes by 2 per cent ($1.70 \to 1.67 \times 10^6\,\mathrm{M}_\odot$ from $\rho_{20} = 0$ to 4) while the stars lose $3 \times 10^5\,\mathrm{M}_\odot$ and remnants and halo gain it together (Figure 18). In the mocks the leading degeneracy is stars against dark mass (remnants plus halo): correlation -0.96, stellar rms $4.7 \times 10^4\,\mathrm{M}_\odot$ inside 10 pc with the amplitude profiled. Pinning the stars removes it (rms 5×10^3). Only then is the remainder a remnants–halo trade (correlation -0.72, rms 1.3–$1.6 \times 10^4\,\mathrm{M}_\odot$, about 1 per cent of the total and comparable to the noise on the total). The pinned ρ_{20} scatter is mostly r_s wandering between 5 and 500 pc at fixed enclosed mass; $M_{\mathrm{DM}}(< 10\,\mathrm{pc})$ is the better-posed quantity.

With the photometric stellar mass imposed the two classes of model separate (Figure 19, right). At $M/N_{19} = 9.0$ without a halo the model is 0.5–1.5 per cent low in the HST dispersions at 1–8 pc (radial PM residual -4.3σ at 3 pc; objective 313.1, $\chi^2_{\mathrm{kin}} = 206.6$), because remnants at $\sim 3\,\mathrm{pc}$ cannot replace missing extended mass. With the halo the objective is 292.5: at fixed photometric stellar mass the halo is preferred by 20.6. The mocks show that the same preference arises on a no-halo truth when the mass per star is underestimated by 16 per cent.

Two representative models are compared component by component in Figure 20: (A) no DM, free stellar mass; (B) DM halo, photometric stellar mass. The mass fractions (stars/remnants/DM) inside 1, 3, 10 and 30 pc are 40/60/–, 60/40/–, 86/14/–, 91/9/– for A and 36/61/3, 50/46/4, 74/20/6, 76/12/12 for B. In both the remnants dominate inside $\sim 1.5\,\mathrm{pc}$; in B the halo is a minor component everywhere inside the data.

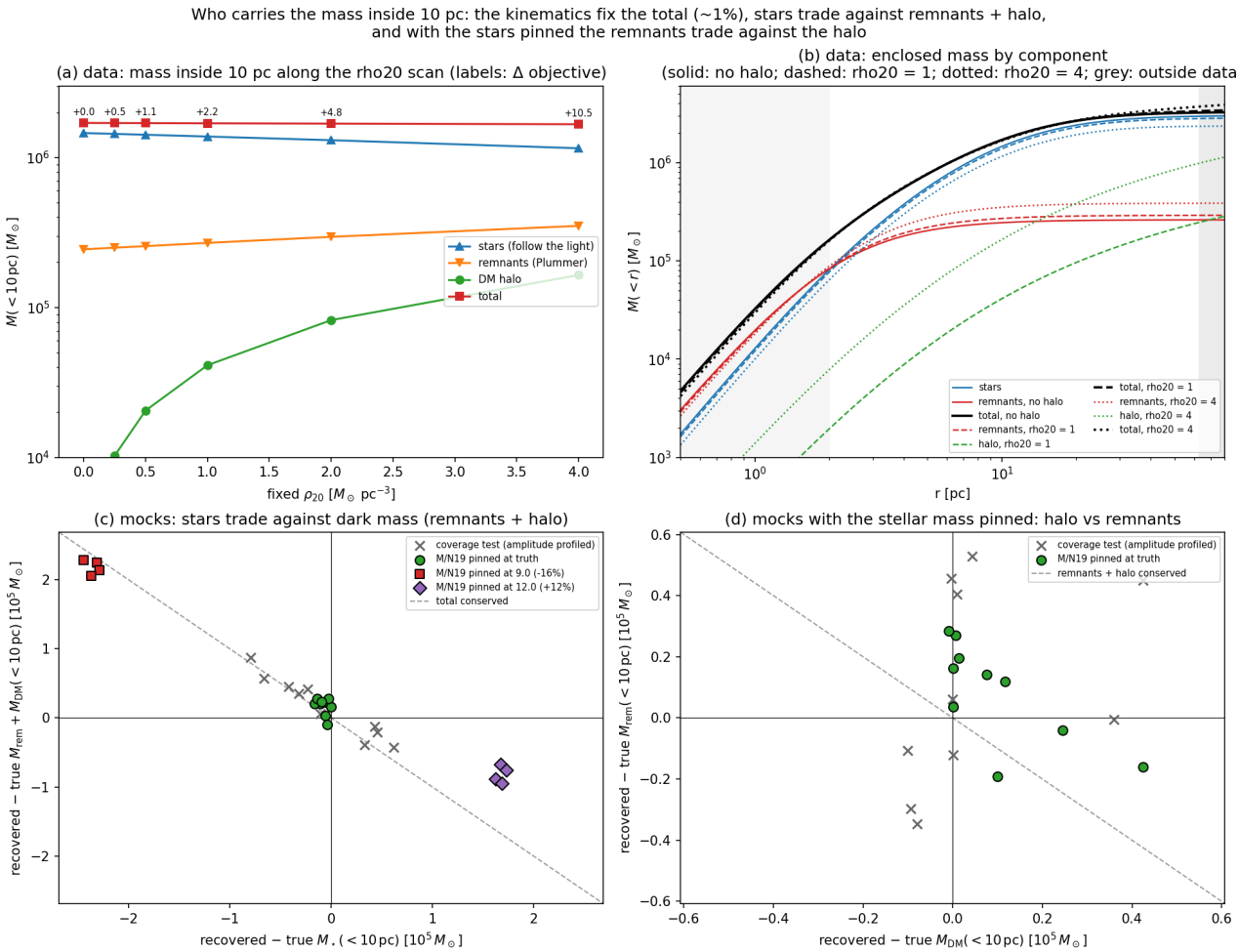


Figure 18: (a) Data: mass inside 10 pc per component along the ρ_{20} scan (labels: Δ objective). (b) Enclosed-mass profiles for $\rho_{20} = 0$, 1 and 4. (c) Mocks: error of the stellar mass against the error of remnants plus halo inside 10 pc. (d) Mocks with the stars pinned: halo against remnant error.

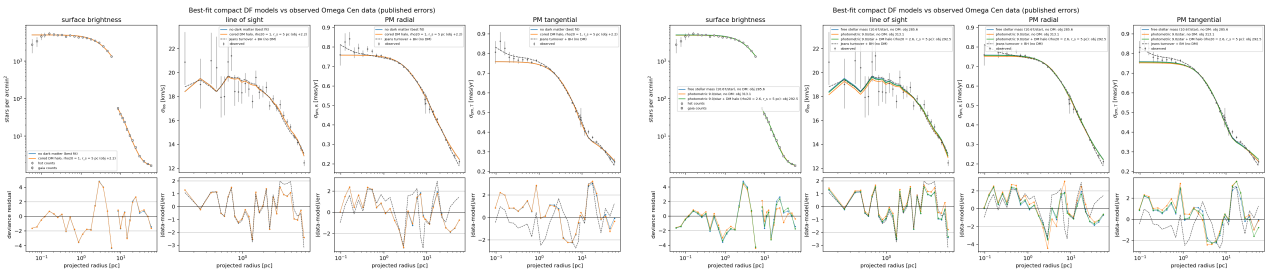


Figure 19: All data with models. Left: the free-mass no-DM fit and the refitted $\rho_{20} = 1$ fit coincide. Right: free-mass no-DM (285.6), photometric $M/N_{19} = 9.0$ without DM (313.1) and with a halo ($\rho_{20} = 2.6$, $r_s = 5$ pc; 292.5).

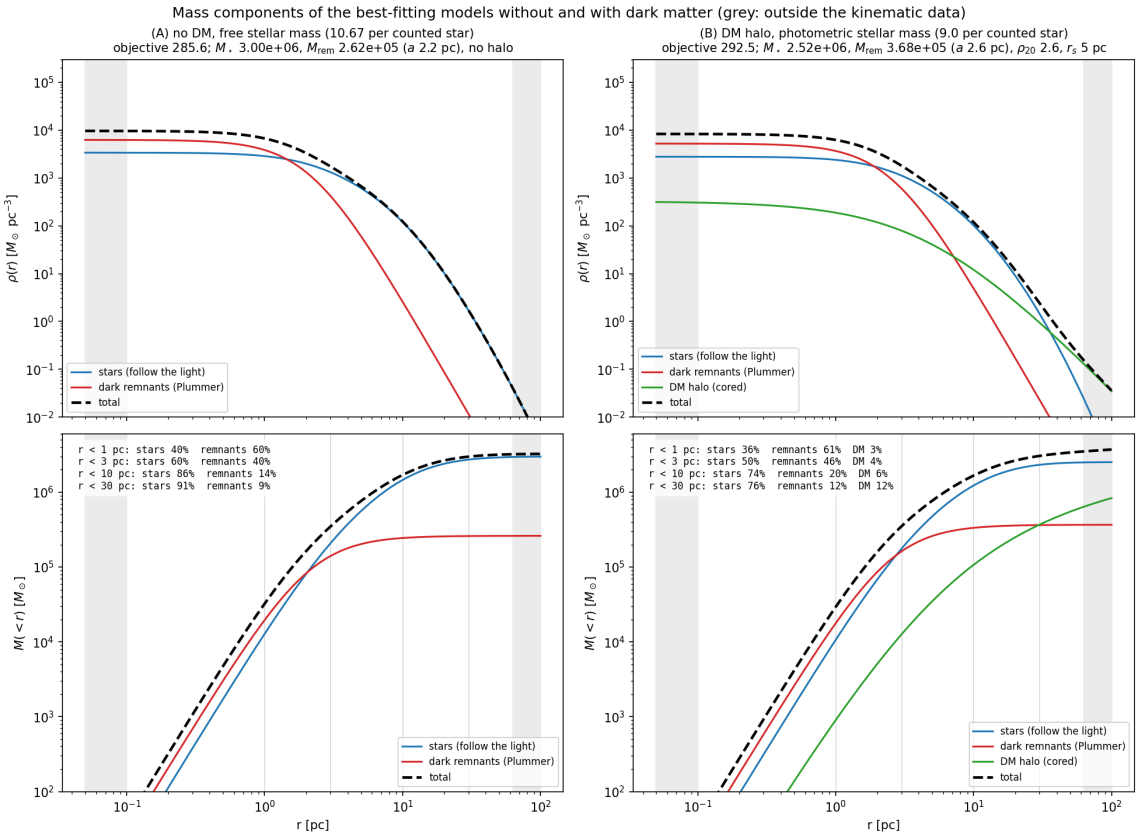


Figure 20: Density (top) and enclosed mass (bottom) of stars, remnants and halo for model A (left) and model B (right). Grey: outside the kinematic data (inside 0.1 pc and beyond 63 pc).

6.7 What can and cannot be said

- **Can:** under this model and the published errors, the total enclosed mass is determined to a few per cent over 2–40 pc, and the best fit with a free stellar mass has no halo. With the stellar mass fixed, the dark mass inside 10 pc follows from the assumed mass per counted star (Table 10).
- **Cannot:** quote a limit on ρ_{20}. With the stellar mass free the mocks give a recovery scatter of 0.5–0.9 $M_\odot\,pc^{-3}$; with it pinned at the truth, still $\sim \pm 0.7$, mostly because r_s is unconstrained. The photometric mass per counted star (8.3–9.8 in the kinematically relevant zones) lies 12–27 per cent below the dynamical 10.67; imposed exactly it implies $\rho_{20} \simeq 2$–3, which is what a 16 per cent mass-per-star error produces on a no-halo mock. A decisive statement needs M/N_{19} to ~ 3 per cent. The likelihood is also not calibrated, rotation is subtracted rather than modelled, and the tracer/mass assumption is known to be only approximate (the mass per bright star varies with radius).

7 N-body tidal disruption of the no-DM and DM models

Models A and B of Figure 20 fit the same data. We asked whether their tidal tails differ in ways that observations could test.

7.1 Set-up

- **Initial conditions** (`bin/nbody/build_ics.py`), 5×10^5 equal-mass particles per model. Stars are sampled from the fitted action DF (AGAMA `GalaxyModel`); remnants and halo from isotropic Eddington DFs in the total potential. A: 459 892 stars and 40 108 remnants (6.5 $M_\odot$

each); B: 268 604 stars, 39 152 remnants and 192 244 halo particles (9.4 $M_\odot$ each; the fitted halo holds 1.81×10^6 $M_\odot$, most of it beyond 60 pc). The sampled cumulative mass matches the model to < 1 per cent at 1–1000 pc; virial ratios 1.0006 and 1.0004.

- **Integration** (`bin/nbody/run_nbody.py`): falcON forces via `pyfalcon` (https://github.com/GalacticDynamics-Oxford/pyfalcon; the PyPI package of that name is unrelated), kick–drift–kick leapfrog, softening 0.3 pc, step 0.01 Myr, static McMillan (2017) potential from AGAMA.
- **Orbit**: ω Cen today from the Baumgardt and Vasiliev (2021) orbit table (https://people.smp.uq.edu.au/HolgerBaumgardt/globular/orbits_table.txt): $(X, Y, Z) = (4.87, -4.07, 1.40)$ kpc, $(U, V, W) = (-95.98, -21.69, -86.82)$ km s^{-1}; pericentre 1.42, apocentre 7.11 kpc in this potential, radial period $\simeq 89$ Myr, retrograde. The clusters start 1955.6 Myr before today (23 pericentres).
- **Flags**: static potential with no dynamical friction or bar (an orbit sub-project will compare these); non-rotating models (the fitted DF is the even part in L_z); two-body relaxation in the simulation is 2.8 times faster than in the real cluster, so the inner ~ 2 pc evolves more than it should, while the envelope and tails ($r > 5$ pc) are collisionless to a few per cent.

Isolation tests (`bin/nbody/analyse_isolation.py`; Figure 21). Over 20 Myr (~ 130 core crossing times) the stellar density stays within 1.8 (A) and 1.5 per cent (B) of the model at 0.3–30 pc, the virial ratio within 0.997–1.001, $|\Delta E/E| \leq 2 \times 10^{-4}$, and σ_r, σ_t and $\beta(r)$ unchanged. The remnant density inside ~ 1 pc drops by 5–8 per cent in the first 5 Myr and then stays constant (readjustment to the softened forces). A step of 0.02 Myr gave $|\Delta E/E| = 8 \times 10^{-4}$ and a 5–13 per cent low central density and was rejected.

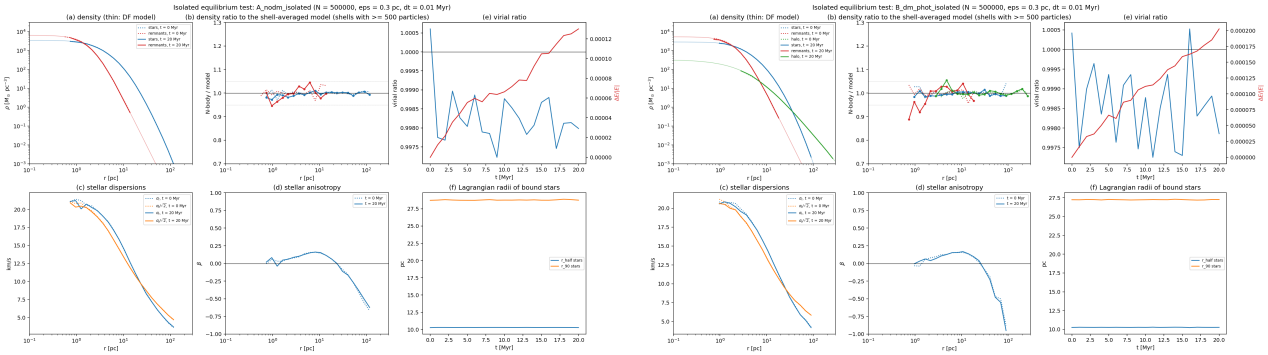


Figure 21: Isolated evolution for 20 Myr of model A (left) and B (right): density per species against the DF model (ratio to the shell-averaged model density), stellar dispersions and anisotropy at 0 and 20 Myr, virial ratio, energy error and Lagrangian radii.

7.2 Evolution and the present-day state

- **Orbit fidelity.** A follows the point-mass orbit to ≤ 71 pc throughout and ends 54 pc and 1.9 km s^{-1} from ω Cen's observed phase-space position. B drifts: 12 pc after 300 Myr, ~ 250 pc after 1.2 Gyr, 566 pc and 15 km s^{-1} today ($b = 8.4°$ instead of 15.0°). This is physical within model B: 1.4×10^6 $M_\odot$ of stripped DM surrounds the orbit and pulls on the cluster.
- **Mass loss** (Figure 22). Today A retains 94.6 per cent of its stars (1.62×10^5 $M_\odot$ stripped) and 99.5 per cent of its remnants; B retains 95.9 per cent of its stars (1.03×10^5 $M_\odot$ stripped), 99.3 per cent of its remnants and 23 per cent of its halo. In the first ~ 500 Myr B's then intact outer halo shielded its stars (stripping rates ~ 5 against 2.5–3 per cent per Gyr); the rates converge after ~ 1.3 Gyr, to 1.95 and 1.81 per cent per Gyr over the last 500 Myr.

- **Tails.** Both models have wrapped their debris into a rosette spanning $\ell = -70°$ to $+70°$, $|b| < 45°$. The spread in energy and L_z of the unbound stars is the same to 2 per cent (3908 vs 3983 $\mathrm{km^2\,s^{-2}}$; 46.2 vs 46.9 $\mathrm{kpc\,km\,s^{-1}}$), and the tail width is the same within errors.

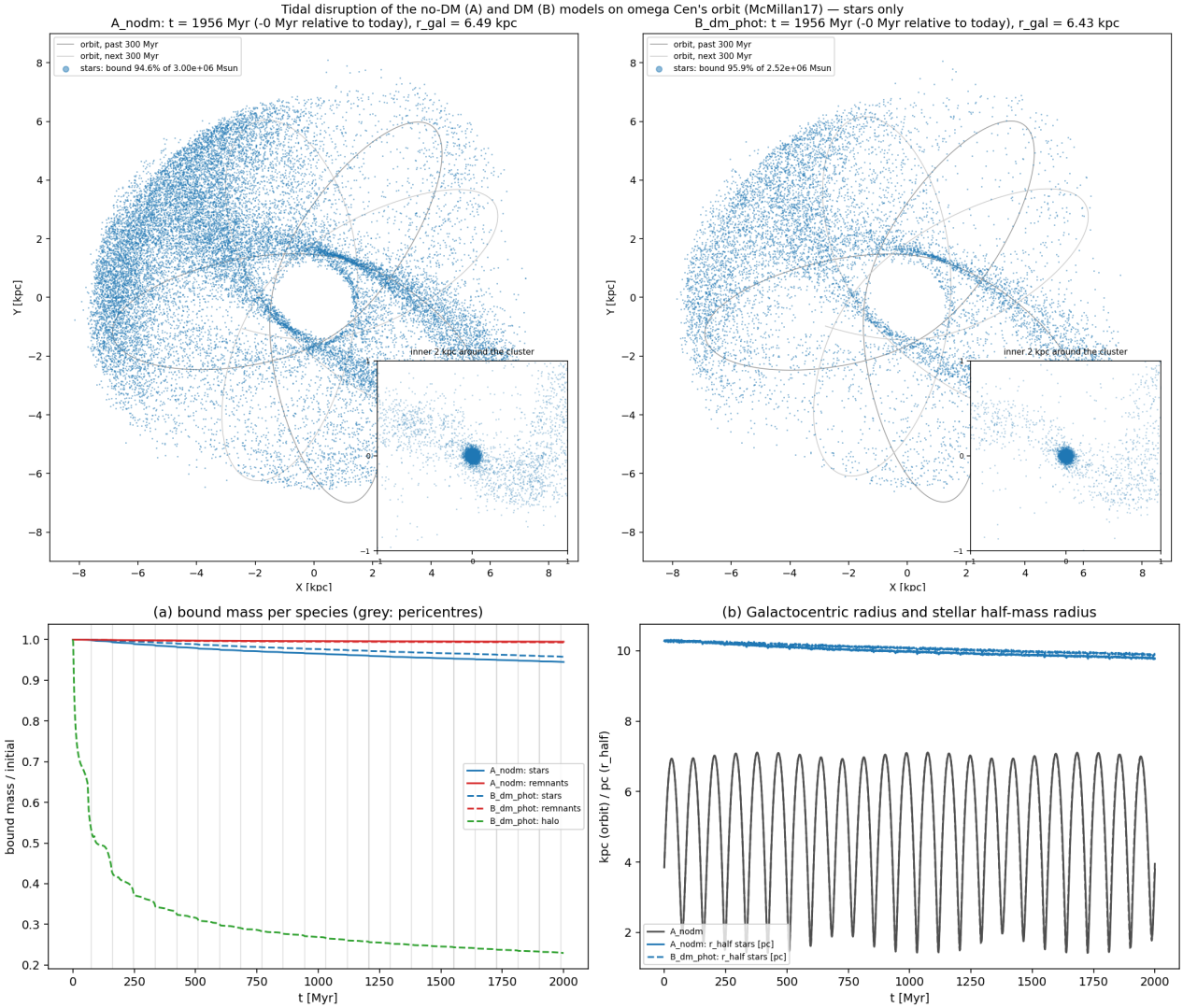


Figure 22: Top: stars of model A (left) and B (right) today in the Galactocentric X–Y plane, with the orbit over ±300 Myr and the inner 2 kpc as an inset. Bottom: bound mass per species against time (grey: pericentres; the present day is $t = 1955.6$ Myr) and the Galactocentric radius and stellar half-mass radius.

7.3 Differences between the two disruptions

Counts are normalized to each model's bound stellar count, because the particle masses differ; sky offsets are relative to each model's cluster (Figures 23 and 24).

- Unbound/bound stars: 5.7 against 4.3 per cent for the whole tail (A/B 1.33); 1.27 within 5° and 10°, 1.09 within 20°. The difference was accumulated in the shielded early phase and depends on the initial halo extent and on the real history.
- Line-of-sight dispersion of tail stars on the same wrap (distance within 0.5 kpc of the cluster) within 10°: 8.5 ± 0.3 against 7.1 ± 0.3 $\mathrm{km\,s^{-1}}$ (A/B 1.2; 1.4 within 5° with 152 and 71 particles).
- Bound cluster: projected dispersion identical to 2 per cent at 1–60 pc (both fit the same data); at 100–200 pc 4.8 against 4.1 $\mathrm{km\,s^{-1}}$.

- Invisible: B's debris is 93 per cent DM by mass; its tails carry three times more remnants per star; the stripped DM displaces B's orbit.

Every luminous difference is ≤ 1.4, the same order as the systematics not modelled (potential, dynamical friction, bar, rotation).

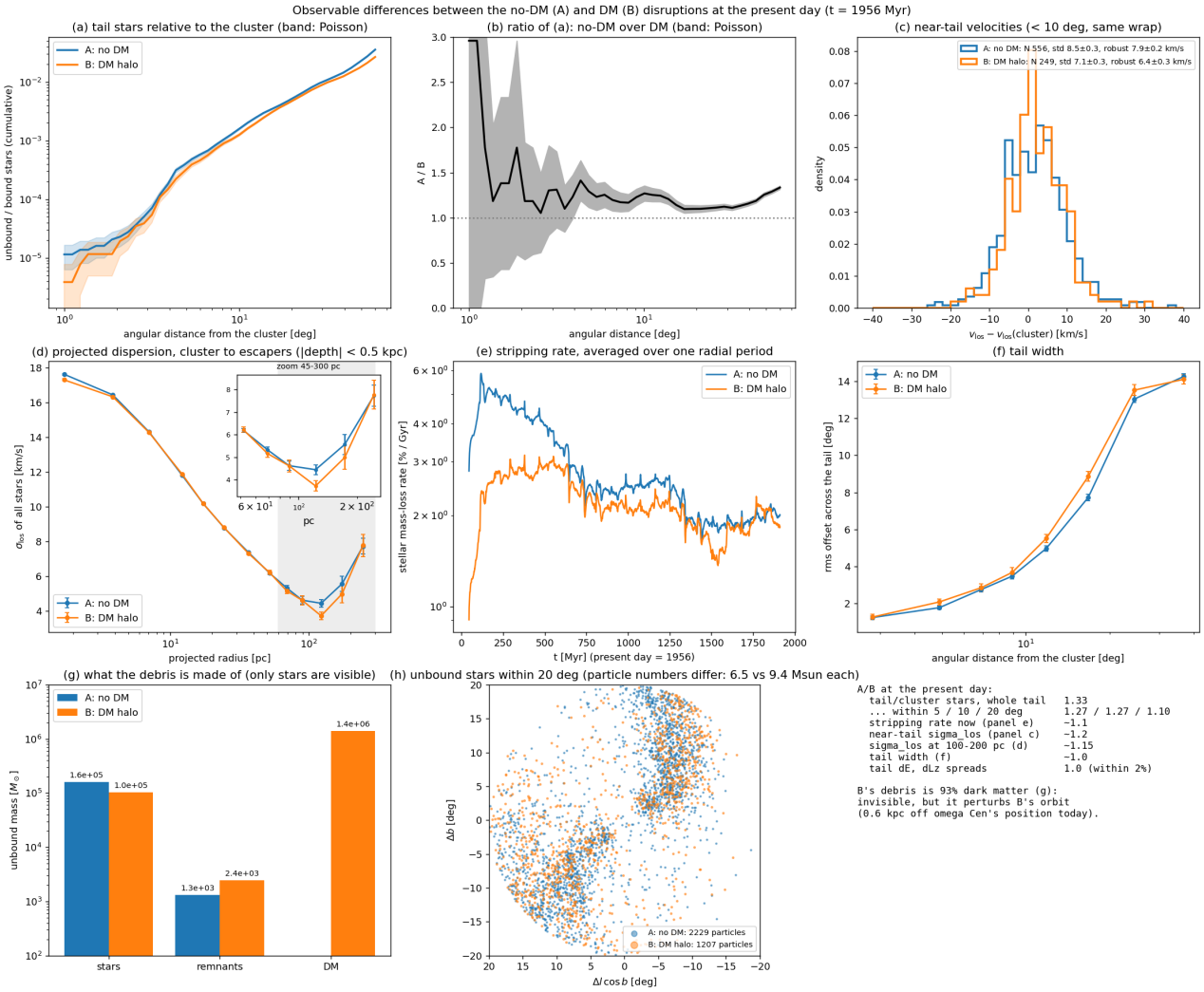


Figure 23: Observable differences today: (a,b) cumulative tail/cluster star ratio and its A/B ratio against angular distance (Poisson bands); (c) near-tail line-of-sight velocities (same wrap, $< 10°$); (d) projected dispersion from the cluster to the escapers; (e) stripping rate per radial period; (f) tail width; (g) debris composition; (h) near-tail sky map; (i) summary of the ratios.

7.4 Comparison with the observed tails

Observed values are those stated in the texts of Kuzma et al. (2021), Kuzma et al. (2026) and Ibata et al. (2019), plus the dispersion of Da Costa (2012) at the tidal radius as quoted by Kuzma et al. (2026) (Figure 25; `bin/nbody/plot_observed_comparison.py`).

- Star-count slope beyond $r_t = 46.4'$ out to 5°: along the tails -2.36 (A) and -2.55 (B) against -3.40 ± 0.20; across -5.51 and -5.27 against -5.22 ± 0.26.
- Stars between the Wilson radius (70.6′) and 5°: 0.102 and 0.100 per cent against ≥ 0.1 per cent (a lower limit); this annulus is dominated by bound outskirts, identical in the two models.
- Tail position angle: 115.4° and 118.8° against $122.9 \pm 2.1°$.

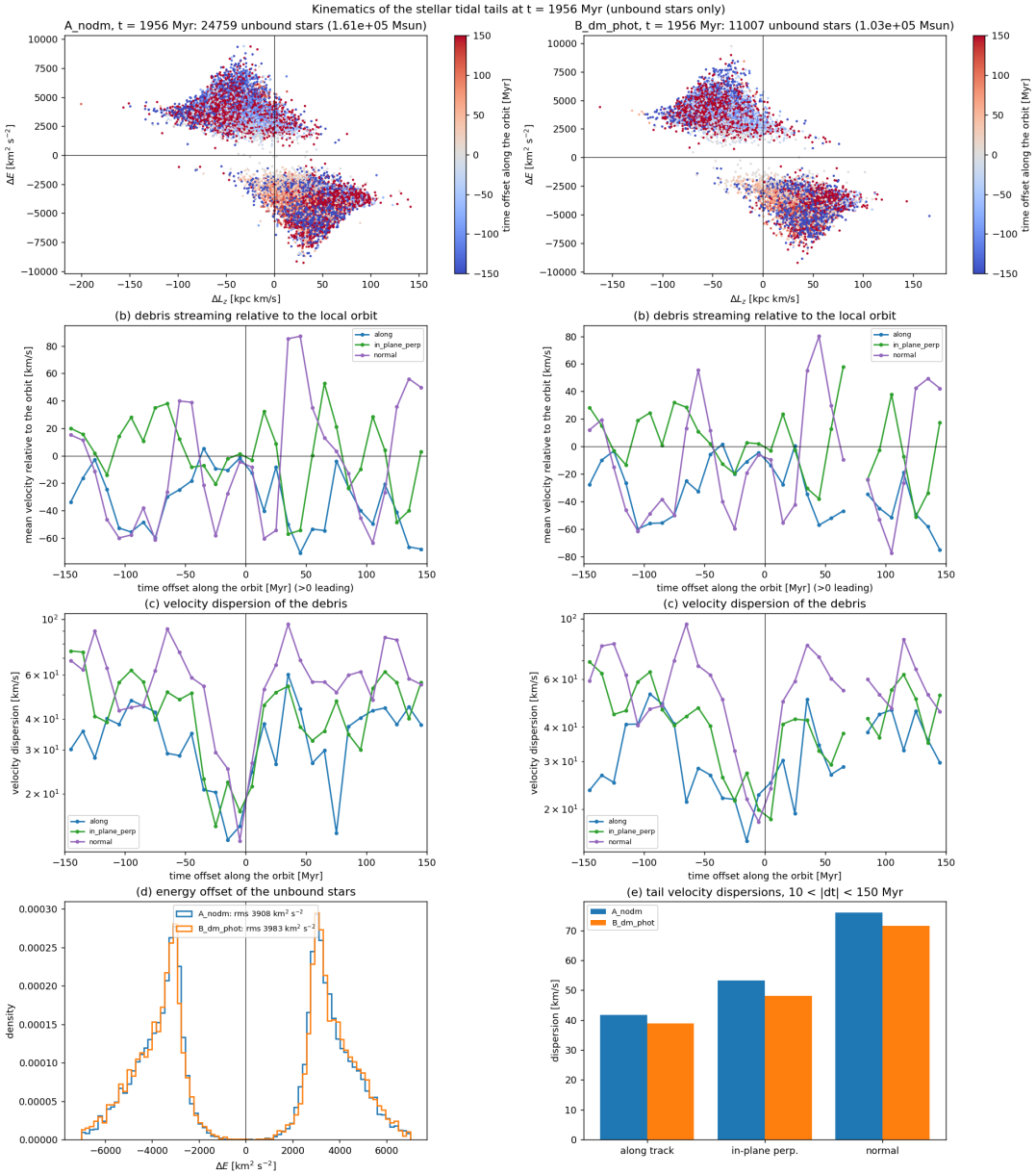


Figure 24: Kinematics of the unbound stars today: energy against L_z offset from the cluster (colour: time offset along the orbit), streaming and dispersions relative to the local orbit along the tails, the ΔE distributions of both models and the tail dispersions per component (these mix orbital wraps within ± 150 Myr and are therefore large).

- Line-of-sight dispersion beyond r_t within 4°: 4.8 km s^{-1} in both, against 6.7 km s^{-1} at r_t; the outermost spectroscopic member, offset by $\sim$ 7 km s^{-1} at 3°, lies within the model spread; neither model has a velocity gradient, as observed.

- Fimbulthul: model debris lies at its position and distance (median 4.2 kpc against 4.1), but it is not Fimbulthul (Figure 26). It is trailing debris stripped at many different times, its velocities spread over 140–255 km s^{-1} (against a mean 199.7 with rms 5.4 km s^{-1} for five stars), and its proper motions ($\mu_\delta = -6$ to -9 mas yr^{-1}) miss the observed members (-9.7 to -11.5); an Ibata-like proper-motion and distance selection keeps no model particle. The difference between the two models' median velocities in that sky box (215 and 202 km s^{-1}) reflects the different age mix of the debris and B's displaced orbit, not a clean DM signature.

The mismatches (profile along the tails too shallow, position angle 4–7° off, no cold Fimbulthul-like stream) are common to both models. They point to the orbit and potential, the integration time (the published Fimbulthul model ran $\sim$ 5 Gyr with a rotating progenitor) and the missing rotation, not to the dark matter.

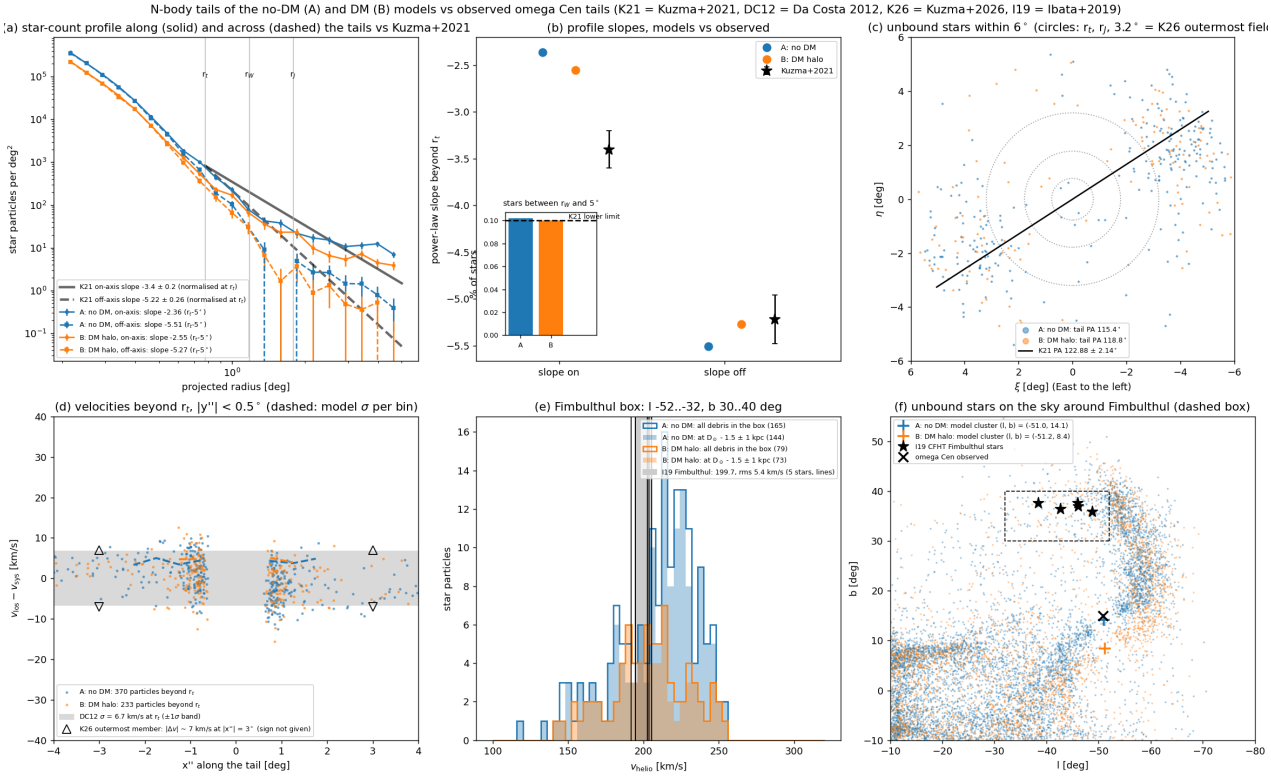


Figure 25: Models against the observed tails: (a,b) star-count profiles along and across the tails and their slopes beyond r_t with the Kuzma et al. (2021) values, and the stellar fraction between the Wilson radius and 5°; (c) unbound stars in the tangent plane with the observed tail direction; (d) line-of-sight velocities beyond r_t with the Da Costa (2012) dispersion and the outermost Kuzma et al. (2026) member; (e,f) the Fimbulthul box ($\ell = -52°$ to $-32°$, $b = 30°$–$40°$) with the five Ibata et al. (2019) velocities. Panel (e) compares unlike populations (see Figure 26).

Testability. The two discriminants that remain, tail counts relative to the cluster (A/B $\simeq$ 1.3) and the near-tail dispersion (A/B $\simeq$ 1.2), are not measured. The only published count is the lower limit above, and equal-mass particles cannot model the preferential escape of low-mass stars, a correction at least as large as the A/B difference. No tail dispersion is published; five spectroscopic members give it to 32 per cent. Separating 8.5 from 7.1 km s^{-1} at 2σ needs $\sim$ 100–200 tail members within 10° with velocities good to $\sim$ 2 km s^{-1}. A sample could be assembled from existing surveys (Gaia DR3 RVS, APOGEE, GALAH, S5, Gaia-ESO), with the same selection applied to the snapshots.

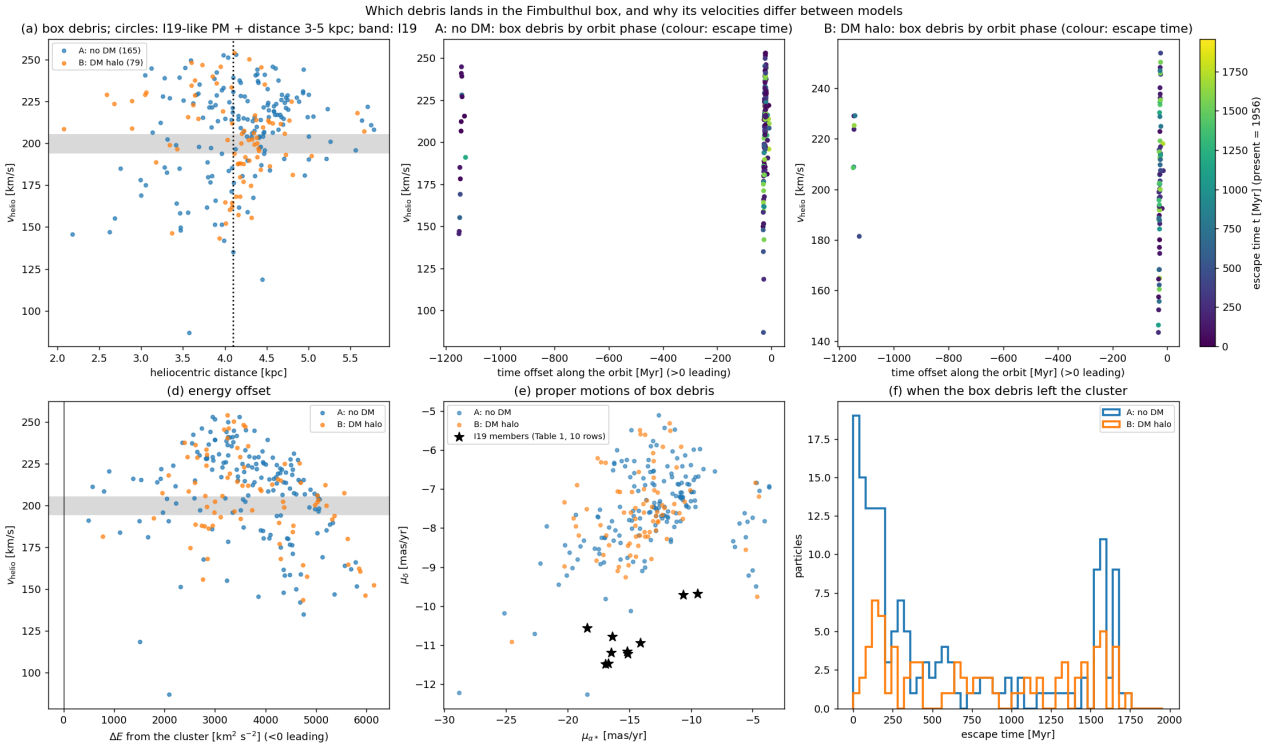


Figure 26: Model debris in the Fimbulthul box: velocity against distance (circles would mark particles passing an Ibata-like selection; there are none), against orbit phase coloured by escape time, and against energy offset; proper motions with the observed members (stars); escape-time distributions.

8 Numerical and software issues found and fixed

- **Optimizer budget and tolerances.** The first recovery batch capped fits at 180 evaluations, about 18 Gauss–Newton steps with nine coordinates. Its relative tolerances were tighter than the evaluation noise. An absolute stopping rule, a 600-evaluation cap and a tenfold tighter equilibrium tolerance fixed both problems.

- **Rejected equilibria in Jacobians.** A rejected probe previously entered as a 10^6 residual. That corrupted the Jacobian and collapsed the trust region to a spurious "converged" stop. Rejected probes are now retried in the opposite direction (the column is zeroed if both fail), and stops close to rejections are flagged. Infeasible starting points are recorded instead of crashing.

- **Projection check range.** A 0.8% fast-vs-direct projection difference occurred only at 80 pc, beyond the data (68 pc). Validation now covers the data radii.

- **Test isolation.** One test module set AGAMA's global units to kpc, so later DF tests refused to run (25 failures and errors in the full suite). A fixture now restores units; the full suite passes (630 passed, 5 skipped).

- **AGAMA time unit.** In kpc–km s^{-1} units AGAMA's time unit is 977.79 Myr, not 1 Gyr. The N-body orbit start used `time=-2.0` and so began 1955.6 Myr before today; the present day is $t = 1955.6$ Myr in those runs (their physics is unaffected). Fixed in `run_nbody.py` (`AGAMA_T_MYR`) and in the orbit tracks of the tail analyses, whose time offsets had been 2.2 per cent off. Verified with a test particle through the same leapfrog and with AGAMA's own orbit (0.5 pc from today's position).

- **N-body centre finder.** Seeded from the previous centre, the shrinking spheres lost the cluster after 1 Myr of orbital motion ($\sim$ 250 pc), corrupting the on-the-fly bound masses; the first launch was stopped after 10 Myr and relaunched. The centre is now predicted from the centre velocity, with a fallback to the median luminous position.

- **Shell densities.** Comparing N-body shell densities with the model density at the shell midpoint biased the ratio by ~ 3 per cent for these steep profiles; the comparison now uses the shell-averaged model density from the enclosed mass.
- **Mass-function completeness.** Zone-dependent faint limits made the mass-function slope flatten outwards spuriously; all zones now use one common limit ($F625W < 21$).
- **Remaining numerical limitation.** In strongly DM-dominated regions ($M_\star \sim 1.4 \times 10^6\,\mathrm{M}_\odot$, $\rho_{20} \sim 8$) the self-consistency iteration can diverge. The normalization tolerance also rejected some extreme starts. Neither affects the minima reported here.

Withdrawn: Gaia error floors

Two diagnostic fits added 0.01 and $0.02\,\mathrm{mas\,yr^{-1}}$ in quadrature to the Gaia errors. They showed that the size of the combined misfit depends strongly on the Gaia errors ($\chi^2_{\rm kin}/N$ falls from 5.12 to 2.09 with $0.02\,\mathrm{mas\,yr^{-1}}$), while the anisotropy conflict persists. Such floors are ad hoc, not a calibrated error model. They have been withdrawn from all model assessment and will not be used. The error-model question is left to a separate, calibrated treatment.

9 Conclusions and next steps

1. The single-transition regularized DF cannot fit the combined ω Cen data. The inner and outer data demand radial and tangential anisotropy respectively. This is established by converged multi-start fits, subset fits and an independent Jeans analysis.
2. The two-transition DF, with no halo and no BH, fits the kinematics better than the best existing Jeans model ($\chi^2_{\rm kin} = 159.2$ vs 190.0 for 89 bins). Its anisotropy follows the Jeans turnover inside $\sim 30\,\mathrm{pc}$, and its DF is smooth. It is a structural near-pass, not a statistical pass.
3. The remaining misfits are localized at the MUSE outer edge and in the 4–20 pc tangential PMs, where the rotation correction is largest. The photometric misfit appears to be dominated by internal scatter in the compilation.
4. Replacing the adopted-error photometry by Poisson star counts leaves the kinematic fit unchanged, removes the artificial outer anisotropy, and exposes a few-per-cent rigidity of the envelope in the HST counts.
5. With the count likelihood and the distance prior, a free halo goes to $\rho_{20} = 0$ and the profile likelihood rises monotonically: $\Delta = +2.2$ at $\rho_{20} = 1$ and $+10.6$ at $4\,\mathrm{M}_\odot\,\mathrm{pc}^{-3}$, for the most compact halo allowed. Twelve realistic mocks recover ρ_{20} with a scatter of 0.5–0.9 (no-halo truths returned up to 1.33; a $\rho_{20} = 1$ truth returned 0), while the total mass profile is recovered within $\pm 5\%$ over 2–40 pc. The observed profile is therefore not a ρ_{20} limit; the data determine the mass profile, not the split.
6. The leading degeneracy is stars against dark mass (remnants plus halo), through the count amplitude. The mass function gives 8.3–9.8 $\mathrm{M}_\odot$ per counted star against the dynamical 10.67. Pinned at the photometric value the data prefer a compact halo by 20.6 in the objective, but mocks show the same preference for a 16 per cent mass-per-star error. With the stars pinned at the truth, ρ_{20} still scatters by $\sim \pm 0.7$ because r_s is free; $M_{\rm DM}(< 10\,\mathrm{pc})$ is the better-posed quantity.
7. In N-body disruption on ω Cen's orbit the no-DM and DM models produce tails that differ by ≤ 30 per cent in star counts relative to the cluster and ~ 20 per cent in near-tail dispersion, with identical energy spreads and widths. Neither quantity is measured yet; the observed profile slope along the tails, the tail position angle and the cold Fimbulthul stream are missed equally by both models.

Objective 1 (fit), proposed. (i) Model rotation with an odd-L_z DF component and fit the mean velocities jointly, instead of subtracting published curves. (ii) Calibrate the Gaia (and HST) error models from literature systematics or from the HST/Gaia overlap, as a separate data task whose method needs approval. (iii) Examine the photometric compilation for internally inconsistent sources. A central BH, separate tracer and mass DFs, and an axisymmetric DF remain later options, to be introduced only when the data demand them.

Objective 2 (DM), proposed. (i) A physical shape separation: DM cannot be mass-segregated, so r_s should not fall below the stellar half-mass radius ($\sim 10\,\mathrm{pc}$), whereas remnants are compact; this removes the compact-halo/remnant degeneracy. (ii) An external prior on $M_\star$ from the present-day mass function measured in the oMEGACat photometry (complete to $F625W \simeq 23$, i.e. $\sim 0.45\,\mathrm{M}_\odot$), with white dwarfs included in the luminous component and NS/BH remnants bounded by retention. (iii) Repeat the mocks with both to measure the remaining scatter. (iv) Intervals only after the likelihood is calibrated (objective 1, item ii). Status: (ii) is done as a sensitivity map (Section 6.5); what is now needed is the error budget of the photometric mass per counted star (the extrapolation below $0.52\,\mathrm{M}_\odot$ and the white-dwarf term) to ~ 3 per cent, and mass-segregated rather than mass-follows-light stars.

N-body, proposed. (i) Iterate model B's starting point so that it ends at ω Cen's present position. (ii) More star particles in the escaping population and a family of B models with different outer-halo extents. (iii) The orbit sub-project: static potential against dynamical friction and bar resonances; rotating DF models. (iv) A tail-member sample with velocities from existing surveys (Gaia DR3 RVS, APOGEE, GALAH, S5, Gaia-ESO), forward-modelled with the same selection on the snapshots.

A Reproducibility

The driver is `bin/run_compact_df_recovery.py` (`prepare-real`, `run`). The figures come from `bin/plot_observed_df_fits.py`, `plot_beta_profiles.py`, `plot_action_coverage.py`, `plot_df_structure.py`, `plot_best_fit_performance.py`, `plot_central_bh_evidence.py`, `plot_count_profiles.py` and `plot_rho20_profile.py`; the count products from `build_count_profiles.py`; the scan and mocks from `run_rho20_profile_scan.sh`, `run_mock_validation.sh` and `run_mock_coverage.sh`. The mass function comes from `bin/build_mass_function.py` (`-feh -1.53` for the Harris variant); the mass-per-star scan and its mocks from `bin/run_mass_per_star_scan.sh` and `bin/run_mock_mass_per_star.sh` (driver option `-mass-per-star ocen_counts_hst_f625w19=M`), plotted by `plot_mass_per_star_profile.py`, `plot_mock_mass_per_star.py`, `plot_mass_budget_tension.py`, `plot_halo_compensation.py`, `plot_remnant_halo_degeneracy.py` and `plot_component_profiles.py`. The N-body tools are in `bin/nbody/`: `build_ics.py`, `run_nbody.py`, `analyse_isolation.py`, `analyse_tails.py`, `plot_snapshot.py`, `plot_tail_kinematics.py`, `plot_discriminants.py`, `plot_observed_comparison.py` and `fimbulthul_box.py`; runs are under `results/nbody/` (present day in `orbit/snap_today.npz`). Plotted data and checksums are in `results/plot_data/`. Markdown records are in `docs/OBSERVED_DF_FLEXIBILITY.md` and `docs/DF_ANISOTROPY_DIAGNOSTICS.md`. Example of the two-transition run:

```
PY=python
$PY bin/run_compact_df_recovery.py prepare-real --out results/df/NAME \
  --two-transition --branches no_halo --n-starts 4 --stop-delta 0.01 \
  --jacobian-seed base --probe-workers 5 --iteration-tolerance 5e-5 \
  --bound stellar.J0=30:800 --bound stellar.J_a=5:500 \
  --bound stellar.b_out=-2:2 --bound matter.r_s=5:500
$PY bin/run_compact_df_recovery.py run --out results/df/NAME \
  --workers 2 --max-calls 600 --detach
```

Table 11: Batches reported here (under `results/df/`). Each batch freezes its source code, configuration, tests and inputs.

Batch	Commit	Content
`observed_df_20260923`	`bd17811`	One transition, default bounds, 2 branches × 2 starts
`observed_df_wide_20260923`	`5486309`	One transition, wide bounds, 2 branches × 6 starts
`dfdiag_{hst_muse,gaia,freedist}_20260923`	`1e246bd`	Diagnostic subset and distance fits
`dftwo_20260923`	`f6ce414`	Two transitions, no halo, 6 starts
`dftwo_halo_20260923`	`bb40db1`	Two transitions + free cored halo, 6 starts (photometric weighting)
`dftwo_counts_20260923`	`36bca6f`	Two transitions, Poisson counts, distance prior, no halo, 6 starts
`rho20free_20260923`, `rho20scan_{0.25,0.5,1,2,4}_20260923`	`f4fd4c7+`	Free halo and the fixed-ρ_{20} profile scan
`mockA_nohalo_20260923`, `mockB_rho1.0_20260923`	`911255e+`	Single-realization mocks (seeds 101, 202)
`cov_A_s{11..15}_20260923`, `cov_B_rho1.0_s{11..15}_20260923`	`f4fd4c7+`	Coverage mocks (one hand start each)
`mpsmock_{truth,9.0,12.0}_{A,B}_s*_20260923`	`7525c77`	Coverage mocks refitted with M/N_{19} pinned
`mpsscan_{8,9,10,11,12}_20260923`	`c3ee6a2`	Real-data scan in M/N_{19} (11, 12 restarted; `*_pass1` kept)
`mpsscan_9_nohalo_20260923`	`b9453de`	$M/N_{19} = 9.0$, no halo
`results/nbody/{A_nodm,B_dm_phot}`	`bcdecb5`	N-body ICs, isolation and tidal runs (present day `orbit/snap_today.npz`)

```
# stellar mass per counted star pinned; N-body ICs and a tidal run
$PY bin/run_compact_df_recovery.py prepare-real ... \
  --mass-per-star ocen_counts_hst_f625w19=9.0
$PY bin/nbody/build_ics.py --out results/nbody/A_nodm --n 500000 --seed 1 \
  --model "results/df/dftwo_counts_20260923::observed_no_halo_start0"
$PY bin/nbody/run_nbody.py --ics results/nbody/A_nodm/ics.npz \
  --out results/nbody/A_nodm/orbit --orbit --tback 1955.584 \
  --eps 0.3 --dt 0.01 --tstop 1955.584 --snap 10 --diag 1
```

References

Jin H. An and N. Wyn Evans. A Cusp Slope-Central Anisotropy Theorem. *ApJ*, 642(2):752–758, May 2006. doi: 10.1086/501040.

H. Baumgardt and E. Vasiliev. Accurate distances to Galactic globular clusters through a combination of Gaia EDR3, HST, and literature data. *MNRAS*, 505:5957–5977, 2021. doi: 10.1093/mnras/stab1474.

James Binney. Distribution functions for spheroids. *MNRAS*, 549(1):stag854, June 2026. doi: 10.1093/mnras/stag854.

Jieun Choi, Aaron Dotter, Charlie Conroy, Matteo Cantiello, Bill Paxton, and Benjamin D. Johnson. MESA Isochrones and Stellar Tracks (MIST). I. Solar-scaled Models. *ApJ*, 823:102, 2016. doi: 10.3847/0004-637X/823/2/102. URL https://arxiv.org/abs/1604.08592.

Aaron Dotter. MESA Isochrones and Stellar Tracks (MIST) 0: Methods for the Construction of Stellar Isochrones. *ApJS*, 222:8, 2016. doi: 10.3847/0067-0049/222/1/8. URL https://arxiv.org/abs/1601.05144.

M. Häberle, N. Neumayer, A. Seth, et al. Fast-moving stars around an intermediate-mass black hole in ω Centauri. *Nature*, 631:285–288, 2024. doi: 10.1038/s41586-024-07511-z.

Maximilian Häberle, N. Neumayer, C. Clontz, A. C. Seth, P. J. Smith, S. Kamann, R. Pechetti, M. S. Nitschai, M. Alfaro-Cuello, H. Baumgardt, A. Bellini, A. Feldmeier-Krause, N. Kacharov, M. Libralato, A. P. Milone, S. O. Souza, G. van de Ven, and Z. Wang. oMEGACat. VI. Analysis of the Overall Kinematics of Omega Centauri in 3D: Velocity Dispersion, Kinematic Distance, Anisotropy, and Energy Equipartition. *ApJ*, 983(2):95, April 2025. doi: 10.3847/1538-4357/adbe67.

Rodrigo Ibata, Michele Bellazzini, Khyati Malhan, Nicolas Martin, and Paolo Bianchini. Identification of the long stellar stream of the prototypical massive globular cluster ω Centauri. *Nature Astronomy*, 3:667–672, 2019. doi: 10.1038/s41550-019-0751-x. URL https://arxiv.org/abs/1902.09544.

P. B. Kuzma, A. M. N. Ferguson, A. L. Varri, and P. Bianchini. Disruption of a giant: spectroscopic identification of members in the periphery and tidal tails of ω Centauri. *MNRAS*, 550:stag1147, 2026. doi: 10.1093/mnras/stag1147. URL https://arxiv.org/abs/2605.23474.

Pete B. Kuzma, Annette M. N. Ferguson, and Jorge Peñarrubia. Detecting globular cluster tidal extensions with Bayesian inference - I. Analysis of ω Centauri with Gaia EDR3. *MNRAS*, 507:1127–1137, 2021. doi: 10.1093/mnras/stab2280. URL https://arxiv.org/abs/2108.02531.

L. Lindegren, S. A. Klioner, J. Hernández, et al. Gaia Early Data Release 3. The astrometric solution. *A&A*, 649:A2, 2021. doi: 10.1051/0004-6361/202039709.

Paul J. McMillan. The mass distribution and gravitational potential of the Milky Way. *MNRAS*, 465:76–94, 2017. doi: 10.1093/mnras/stw2759. URL https://arxiv.org/abs/1608.00971.

Renuka Pechetti, Sebastian Kamann, Davor Krajnović, Anil Seth, Glenn van de Ven, Nadine Neumayer, Stefan Dreizler, Peter M. Weilbacher, Sven Martens, and Florence Wragg. ω Centauri: a MUSE discovery of a counter-rotating core. *MNRAS*, 528(3):4941–4957, March 2024. doi: 10.1093/mnras/stae294.

S. C. Trager, I. R. King, and S. Djorgovski. Catalogue of Galactic globular-cluster surface-brightness profiles. *AJ*, 109:218, 1995.

Eugene Vasiliev. AGAMA: action-based galaxy modelling architecture. *MNRAS*, 482(2):1525–1544, January 2019. doi: 10.1093/mnras/sty2672.

Eugene Vasiliev and Holger Baumgardt. Gaia EDR3 view on galactic globular clusters. *MNRAS*, 505 (4):5978–6002, August 2021. doi: 10.1093/mnras/stab1475.